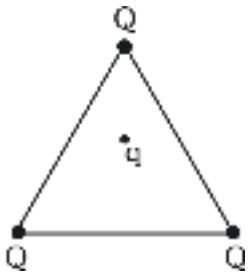
PHYSICS

1) The ratio of q and Q so as to make the system in equilibrium is :



(1) 1 : 3

(2) $1:-\sqrt{3}$

(3) $1:\sqrt{3}$

(4) None of these

2) Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and the other is reduced by 10%. The new force of repulsion at the same distance would be

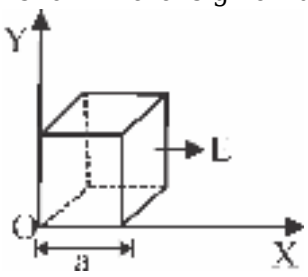
(1) 100 N

(2) 121 N

(3) 99 N

(4) None of these

3) The electric field in a region shown here is given by $\vec{E} = E_0 x^{\frac{3}{2}}\hat{i}$ volt/m. Then the total electric flux through the cube of side a is :



(1) Zero

(2) E_0

(3) $E_0a^{5/2}$

(4) $E_0a^{7/2}$

4)

A thin disc of radius R is charged on its surface with surface charge density $\sigma = \sigma_0 r$, here σ_0 is constant and r is distance from the center. Find total charge on the disc

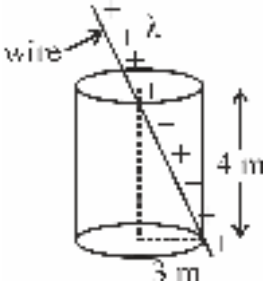
(1) $\frac{2\pi\sigma_0 R^3}{3}$

(2) $\pi\sigma_0 R^3$

(3) $3\pi\sigma_0 R^3$

(4) $\frac{4}{3}\pi\sigma^0 R^3$

5) Linear charge density of wire is 8.85 μC/m. Radius and height of the cylinder are 3m and 4 m.



Then find the flux passing through the cylinder :-

(1) 5×10^6 volt-m

(2) 3×10^6 volt-m

(3) 4×10^6 volt-m

(4) None

6) The electric field is 100 V/m, at a distance of 20 cm from the centre of a dielectric sphere of radius 10 cm. Then E at 3 cm distance from the centre of sphere is :–

(1) 100 V/m

(2) 125 V/m

(3) 120 V/m

(4) zero

7) In a certain region of space, the potential is given by : $V = k[2x^2 - 2y^2 + z^2]$. The electric field at the point (1, 1, 1) has magnitude.

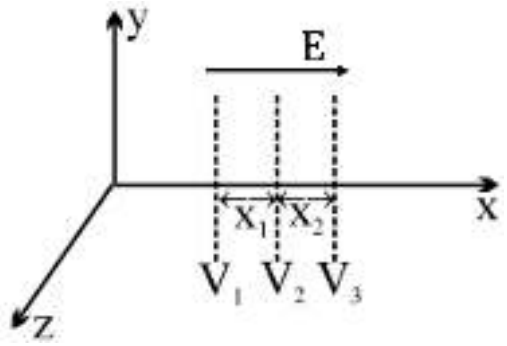
(1) 6k

(2) $2k\sqrt{6}$

(3) $2k\sqrt{3}$

(4) $4k\sqrt{3}$

8) In an electric field shown in figure three equipotential surfaces are shown. If function of electric field is $E = 2x^2 V/m$, and given that $V_1 - V_2 = V_2 - V_3$, then we have



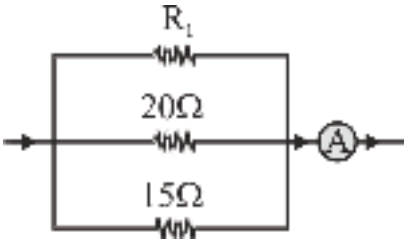
(1) $x_1 = x_2$

(2) $x_1 > x_2$

(3) $x_2 > x_1$

(4) data insufficient

9) In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammeter reads 0.8 ampere. What is the value of R_1 ?



(1) 30 ohms

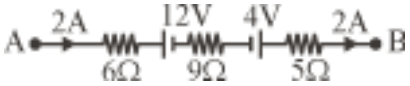
(2) 40 ohms

(3) 50 ohms

(4) 60 ohms

10)

For given circuit find $V_A - V_B$:-



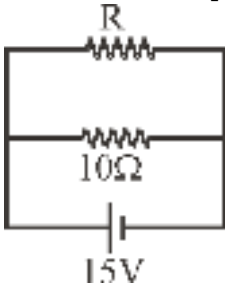
(1) 24 V

(2) 14 V

(3) 32 V

(4) 48 V

11) Power dissipated in the circuit shown in the figure is 67.5 watt .The value of R is :-



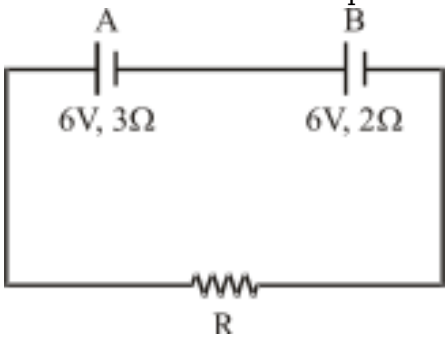
(1) 5 Ω

(2) 15 Ω

(3) 10 Ω

(4) 20 Ω

12) Two sources of emf 6 V and internal resistance 3 Ω and 2 Ω are connected to an external resistance R as shown. If potential difference across battery A is zero, then value of R is :



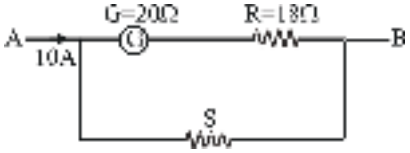
(1) 1 Ω

(2) 2 Ω

(3) 3 Ω

(4) 4 Ω

13) Full scale deflection current for galvanometer is 1 Amp. When shunt 'S' is connected then, what should be the value of shunt resistance so that galvanometer shows half scale deflection.



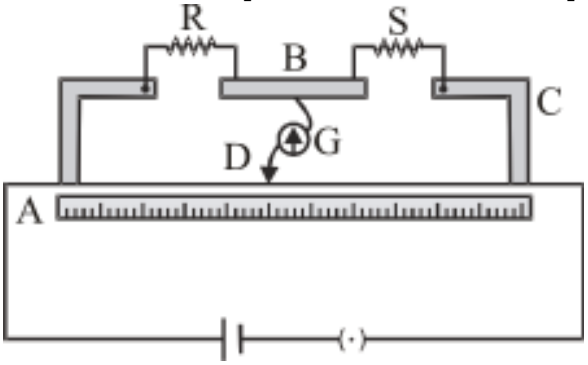
(1) 10 Ω

(2) 1 Ω

(3) 12 Ω

(4) 2 Ω

14) Circuit diagram of metre bridge is shown in figure. The null point is found at a distance of 40 cm from A. If now a resistance of 12Ω is connected in parallel with S, the null point occurs at 64 cm.



Determine the value of R and S.

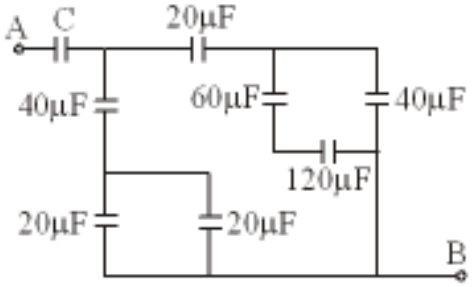
(1) $\frac{40}{3}\Omega$, 20Ω

(2) $\frac{10}{3}\Omega$, 10Ω

(3) 11Ω, 22Ω

(4) $\frac{80}{3}\Omega$, 40Ω

15) Figure shows a network of capacitors. Find the value of C, if the effective capacitance between the points A and B is 12 μF.



(1) 18 μF

(2) 20 μF

(3) 10 μF

(4) 5 μF

16) A slab of copper of thickness $\frac{d}{3}$ is introduced between the plates of a parallel plate capacitor where d is the separation between its two plates. If the capacitance of the capacitor without the copper slab is C and with copper slab is C' then $\frac{C'}{C}$ is :-

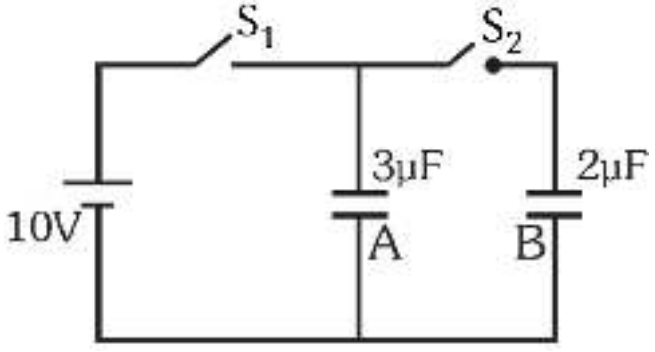
(1) $\sqrt{2}$

(2) 2

(3) $\frac{3}{2}$

(4) $\frac{1}{\sqrt{2}}$

17) In given circuit switch S_1 is closed and S_2 is open. After long time S_1 is opened and S_2 is closed. Calculate charge on each capacitor. (Initially both capacitors are uncharged)



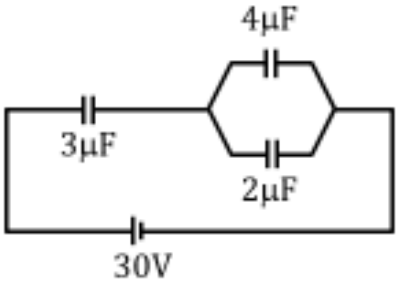
(1) 18μC, 12μC

(2) 15μC, 15μC

(3) 10μC, 30μC

(4) 5μC, 25μC

18) In the circuit shown in figure, match the following two columns.



Column-I	**Column-II**
(I) Charge given by battery	(a) 20 μC
(II) Charge of 4μF capacitor	(b) 30 μC
(III) Charge of 2 μF capacitor	(c) 60 μC
(IV) Charge of 3 μF capacitor	(d) 40 μC

(1) I → a ; II → b ; III → b ; IV → c

(2) I → c ; II → d ; III → a ; IV → c

(3) I → b ; II → a ; III → c ; IV → d

(4) I → c ; II → d ; III → c ; IV → a

19) 25 W, 200 V and 100 W, 200 V bulbs are connected in series to a source of 400 volts. Which bulb will fuse ?

(1) 25 W

(2) 100 W

(3) Both will fuse at the same time

(4) None of the bulbs will fuse

20) A parallel plate capacitor is charged using a battery. After the capacitor is fully charged, the battery is disconnected and the plates are moved apart, then :-

(1) the charge on the plates increases

(2) the potential decreases

(3) the capacitance increases

(4) the electric fields strength remains constant.

21) Two identical metal plates are given positive charges Q_1 and Q_2 ($< Q_1$) respectively. If they are brought close together to form a parallel plate capacitor with capacitance C, the potential difference between them is :-

(1) $\frac{Q_1 + Q_2}{2C}$

(2) $\frac{Q_1 + Q_2}{C}$

(3) $\frac{Q_1 - Q_2}{C}$

(4) $\frac{Q_1 - Q_2}{2C}$

22) A parallel plate capacitor of value 1.77 μF is to be designed using a dielectric material of dielectric constant = 200, break through strength = 3×10^6 Vm^{-1}. In order to make such a capacitor, which can withstand a potential difference of 20V across the plates, the separation between the plates d and the area of the plates A(each) can be :

(1) d = 10^{-6} m, A = 10^{-3} m^2

(2) d = 10^{-5} m, A = 10^{-2} m^2

(3) d = 10^{-4} m, A = 10^{-4} m^2

(4) d = 10^{-4} m, A = 10^{-5} m^2

23) The position (x) of a particle at any time (t) is given by
$x(t) = 4t^3 - 3t^2 + 2$
The acceleration and velocity of the particle at time t = 2 sec are respectively:

(1) 16 ms^{-2} and 22 ms^{-1}

(2) 42 ms^{-2} and 36 ms^{-1}

(3) 48 ms^{-2} and 36 ms^{-1}

(4) 12 ms^{-2} and 25 ms^{-1}

24) A ball is projected vertically upward, if ball passes from same height at t = 2 and 10 sec, after projection, maximum height and time of flight of ball in this motion are x and y respectively, then x/y = ?

(1) 30

(2) 15

(3) 7.5

(4) none

25) Two trains A and B, 100 m and 60m long, are moving in opposite directions on parallel tracks. The velocity of shorter train is 3 times of that of the longer one. If the trains take 4s to cross each other, the velocities of the trains are:

(1) v_A = 10 m/s, v_B = 30 m/s

(2) v_A = 2.5 m/s, v_B = 7.5 m/s

(3) v_A = 20 m/s, v_B = 60 m/s

(4) v_A = 5 m/s, v_B = 15 m/s

26) A particle is moving as x = t and $y = t^2$ then (All units are in S.l. system):

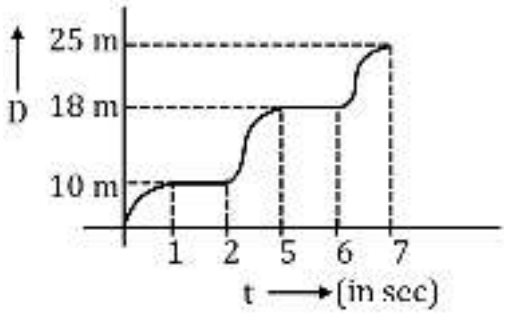
(1) Its path equation is $y = x^2$

(2) motion is accelerated with $\vec{a} = 2\hat{j}$

(3) displacement of particle at t = 1 is $\sqrt{2}$

(4) all are correct

27) A man is moving on a straight road if its distance versus time curve is shown, then about its



displacement upto 7 sec which is correct:

(1) its displacement may be 9 m

(2) its displacement may be 11 m

(3) its displacement may be 25 m

(4) all are possible

28) Velocity and acceleration of a particle at some instant of time are $v = (3\hat{i} + 4\hat{j})ms^{-1}$ and $a = (-6\hat{i} + 8\hat{j})ms^{-2}$ respectively. At the same instant particle is at origin. Maximum x-coordinate of particle will be :

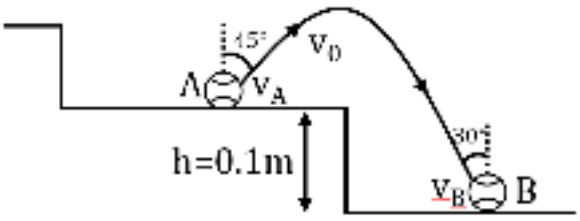
(1) 1.5 m

(2) 0.75 m

(3) 2.25 m

(4) 4 m

29) A ball is dropped onto a step at point A and rebounds with a velocity v_0 at an angle of 45° with the vertical. The value of v_0 knowing that just before the ball bounces at point B its velocity v_B forms



an angle of 30° with the vertical is :

(1) 0.7 m/s

(2) 1.4 m/s

(3) 2.1 m/s

(4) 2.8 m/s

30) **In this type of questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :**

Assertion: If dot product and cross product of $\vec{P}$ and $\vec{Q}$ are zero, it implies that one of the $\vec{P}$

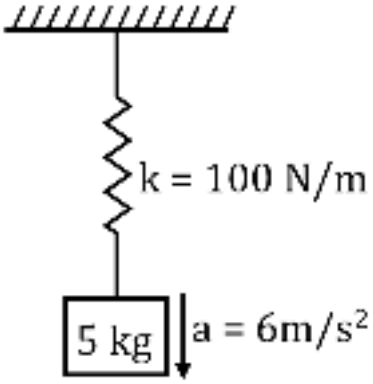
vector and $\vec{Q}$ must be null vector.
Reason: If one of them either $\vec{P}$ or $\vec{Q}$ is null vector then product will be zero.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason if false.
(4) If both assertion and reason are false.

31) An open water tight railway wagon of mass 5×10^3 kg coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be : -

(1) 0.5 m/s
(2) 2 m/s
(3) 1 m/s
(4) 1.5 m/s

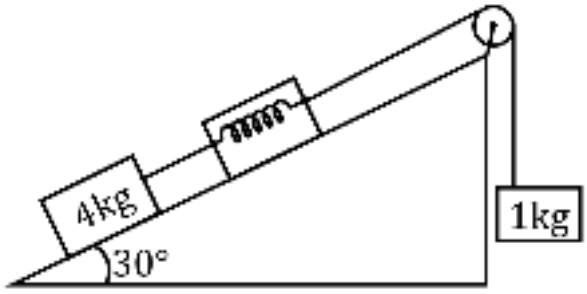
32) A 5 kg block is hang with spring as shown, if acceleration of block at any instant is a = 6 m/s^2,



then elongation in spring at this instant will be :

(1) 50 cm
(2) 30 cm
(3) 20 cm
(4) cannot be calculated

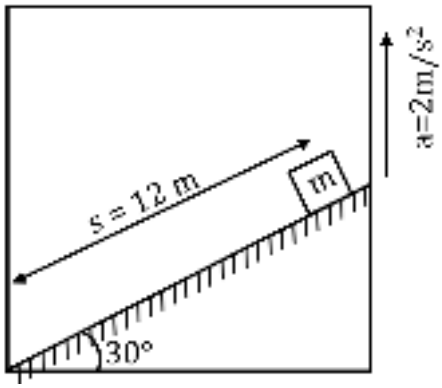
33) In the given arrangement final reading of spring balance ?



(1) 12 N
(2) 10 N
(3) 16 N

(4) None

34) A block of mass m is placed at the top of inclined plane having angle of inclination 30° as shown lift is moving with upward acceleration a =2 m/s^2 then, time taken by particle to reach at bottom of inclined plane will be :



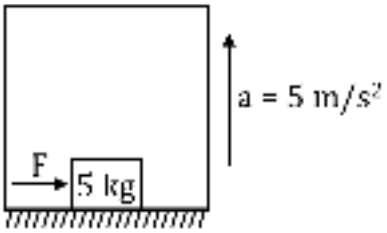
(1) 2 sec

(2) 1 sec

(3) 4 sec

(4) None

35) If coefficient of friction between block and lift is 0.2. Find minimum force F to move the block.



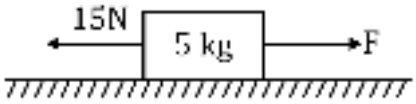
(1) 25 N

(2) 20 N

(3) 15 N

(4) 10N

36) A block of mass 5kg is placed on a horizontal surface with coefficient of friction 'μ = 0.2', then the maximum and minimum value of force F for which the the block remains at rest (g = 10 m/s^2);



(1) 15N, 10 N

(2) 25N, 5 N

(3) 10N, 25N

(4) 5N, 10N

37) A cricket ball of mass 150 g has an initial velocity $u = \left(3\hat{i} + 4\hat{j}\right) ms^{-1}$ and a final velocity $v = \left(-3\hat{i} - 4\hat{j}\right) ms^{-1}$ after being hit. The change in momentum is: (in kg ms^{-1})

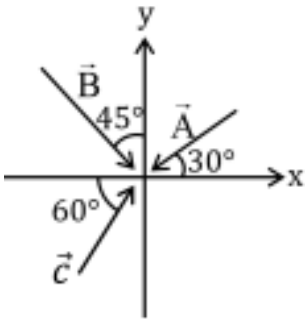
(1) zero

(2) $-\left(0.45\hat{i}+0.6\hat{j}\right)$

(3) $-\left(0.9\hat{i}+1.2\hat{j}\right)$

(4) $-5\left(\hat{i}+\hat{j}\right)$

38) In the given figure vector $\vec{A}, \vec{B}$ and $\vec{C}$ are shown. Find angle between given vectors : -



(i) $\vec{A}$ and $\vec{B}$

(ii) $\vec{A}$ and $\vec{C}$

(iii) $\vec{B}$ and $\vec{C}$

(1) 105°, 150°, 150°

(2) 105°, 150°, 105°

(3) 150°, 150°, 150°

(4) 105°, 105°, 105°

39) The position of a body moving with a non-uniform speed is given by $x = 4t + 3t^2$, then 4 and 3 in equation are:

(1) dimensionless constant

(2) dimensionless variable

(3) dimensional constant

(4) dimensional variable

40) A physical quantity x is given by $x = \frac{2k^3\ell^2}{m\sqrt{n}}$. The percentage error in the measurements of k, □, m and n are 1%, 2%, 3% and 4% respectively. The value of x is uncertain by:

(1) 8%

(2) 10%

(3) 12%

(4) none of these

41) One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 10 divisions of vernier scale coincide with 8 small divisions of the main scale, the least count of the callipers is:

(1) 0.01 cm

(2) 0.02 cm

(3) 0.05 cm

(4) 0.005 cm

42) Equation of trajectory for a projectile is given by $y = x - \frac{x^2}{2}$, then match column I to column II for this projectile & select correct option.

Column-I		Column-II	
(A)	Horizontal Range	(P)	$\frac{1}{2}$ m
(B)	Maximum height	(Q)	1 m
(C)	Angle of projection	(R)	2 m
(D)	Angle of strike	(S)	45°
		(T)	$\tan^{-1}\left(\frac{1}{2}\right)$

(1) A-P ; B-P ; C-S ; D-S

(2) A-R ; B-Q ; C-S ; D-S

(3) A-R ; B-P ; C-T ; D-T

(4) A-R ; B-P ; C-S ; D-S

43) A gardener waters the plants by a pipe of diameter 1 mm. The water comes out at the rate of 10 cm^3/sec. The reactionary force exerted on the hand of the gardener is :

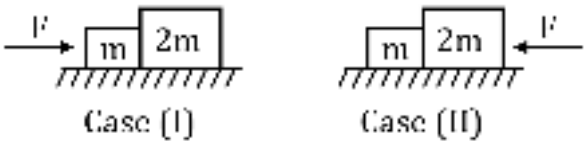
(1) zero

(2) 1.27×10^{-2} N

(3) 1.27×10^{-4} N

(4) 0.127 N

44) Two blocks are in contact on a frictionless table. One applied from left and one from right on masses. In the two cases respectively, the ratio of contact force between the blocks will be : -



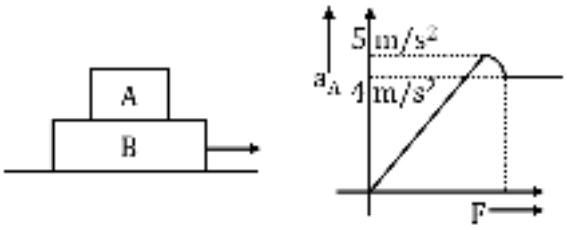
(1) 1 : 4

(2) 2 : 1

(3) 1 : 1

(4) 2 : 3

45) Acceleration of upper block A vs F curve is shown in diagram. If coefficient of static and kinetic friction between block A and B is μ_s and μ_k, then value of ($\mu_k + \mu_s$) will be :



(1) 0.4

(2) 0.5

(3) 0.9

(4) 0.7

CHEMISTRY

1) At same temperature and pressure, three flasks of equal volume contain He, H_2 and O_3 gas respectively. Find the incorrect statement.

(1) Number of moles of gases in three flasks will be same.

(2) Number of molecules in three flasks will be same.

(3) Number of atoms in three flasks will be same.

(4) Weight of gases present in all the three flask will be different.

2) 4.4 gm of CO_2 and 2.24 litre of H_2 at STP are mixed in a container. The total number of molecules present in the container will be -

(1) 6.022×10^{23}

(2) 1.2044×10^{23}

(3) 2 moles

(4) 6.023×10^{24}

3) A compound contains 54.55% carbon, 9.09% hydrogen and 36.36% oxygen. The empirical formula of this compound is-

(1) C_3H_5O

(2) $C_4H_8O_2$

(3) $C_2H_4O_2$

(4) C_2H_4O

4) Which of the following has the maximum number of atoms ?

(1) 28 g of $N_2(g)$

(2) 44.8 L of $O_2(g)$ at STP

(3) 18 mL of $H_2O(l)$

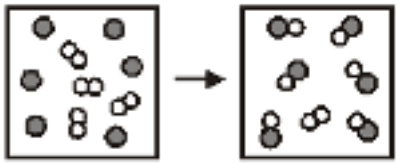
(4) 1.5 mole of He(g)

5) **Assertion :** A compound contains 3.2% oxygen by mass, then minimum molecular mass of the

compound is 500.
Reason : For minimum molecular mass of the compound, it must have one atom of that element.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

6) The following diagram represents the reaction of A_2 (unshaded spheres) with B (shaded spheres). What is the balanced chemical equation for this reaction, and what is the limiting reactant?



(1) $A_2 + 2B \rightarrow 1AB$; A_2 is the limiting reactant
(2) $A_2 + 2B \rightarrow 2AB$; B is the limiting reactant
(3) $4A_2 + 6B \rightarrow 6AB$; A_2 is the limiting reactant
(4) $4A_2 + 6B \rightarrow 6AB$; B is the limiting reactant

7) If there are five energy states in case of H-atom, then the number of lines in UV region would be:

(1) 3
(2) 4
(3) 2
(4) zero

8) Which is correct for any H like species :-

(1) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$
(2) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
(3) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$
(4) $(E_2 - E_1) = \frac{1}{4}(E_3 - E_2) = \frac{1}{9}(E_4 - E_3)$

9) What is the shortest wavelength line in the Paschen series of Li^{+2} ion ?

(1) $\frac{R}{9}$
(2) $\frac{9}{R}$
(3) $\frac{1}{R}$
(4) $\frac{9R}{4}$

10) The potential energy of an electron in hydrogen atom is –6.8 eV. Indicate in which excited state

electron is present :

(1) First

(2) Second

(3) Third

(4) Forth

11) The orbital diagram in which the Aufbau principle is violated is :

(1) 

(2) 

(3) 

(4) 

12) Match the columns

Column-I (Rules)		Column-II (Statements)	
(A)	Hund's Rule	(p)	No two electrons in an atom can have the same set of four quantum numbers
(B)	Aufbau Principle	(q)	Half-filled and completely filled orbitals have extra stability.
(C)	Pauli Exclusion Principle	(r)	Pairing of electrons in the Principle orbitals belonging to the same subshell does not take place until each orbital is singly occupied.
(D)	Heisenberg's Uncertainty Principle	(s)	It is impossible to determine the exact position and exact momentum of a subatomic particle simultaneously
		(t)	In the ground state of atoms, orbitals are filled in the order of their increasing energies.

(1) A – (r), B – (p), C – (t), D – (s)

(2) A – (r), B – (t), C – (s), D – (p)

(3) A – (r), B – (t), C – (p), D – (s)

(4) A – (t), B – (r), C – (p), D – (s)

13) The equilibrium constant of the following are :

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) K_1$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g) K_2$

$H_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons H_2O(g) K_3$

The equilibrium constant (K) of the reaction,

$2NH_3(g) + \frac{5}{2} O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$ will be :

(1) $\frac{K_2K_3^3}{K_1}$

(2) $\frac{K_2K_3}{K_1}$

(3) $\frac{K_2^3K_3}{K_1}$

(4) $\frac{K_1K_3^3}{K_2}$

14) For the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, the value of K_c at 800° C is 0.1. When the equilibrium concentrations of both the reactants is 0.5 mol, what is the value of K_p at the same temperature?

(1) 0.5

(2) 0.1

(3) 0.01

(4) 0.025

15) For the reaction, $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$ the value of K_p/K_c is equal to:

(1) 1.0

(2) RT

(3) $\sqrt{RT}$

(4) $\frac{1}{RT}$

16) Which of the following cannot act as both Bronsted acid and base.

(1) HS^-

(2) CO_3^{-2}

(3) HCO_3^-

(4) H_2O

17) Solubility product constants (K_{sp}) of salts of types MX, MX_2 at M_3X at temperature T are 4.0×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15} respectively. Solubility (mol dm^{-3}) of the salt at temperature T are in the order

(1) $MX > MX_2 > M_3X$

(2) $M_3X > MX_2 > MX$

(3) $MX_2 > M_3X > MX$

(4) $MX > M_3X > MX_2$

18) What is the minimum volume of water required to dissolve 1 g of calcium sulphate at 298 K. K_{sp} for $CaSO_4$ is 9.0×10^{-6}.

(1) 2.45 L

(2) 4.08 L

(3) 4.90 L

(4) 3.00 L

19) The ratio of volumes of CH_3COOH, 0.1 N to CH_3COONa, 0.1 N required to prepare a buffer solution of pH 5.74 is (given : pK_a of CH_3COOH = 4.74)

(1) 10 : 1

(2) 5 : 1

(3) 1 : 5

(4) 1 : 10

20) The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of first order reaction is -5×10^3 K. The value of E_a of the reaction is. Choose the correct option for your answer.
[Given R = 8.314 JK^{-1} mol^{-1}]

(1) 41.5 kJ mol^{-1}

(2) 83.0 kJ mol^{-1}

(3) 166 kJ mol^{-1}

(4) –83 kJ mol^{-1}

21) For the reaction : $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ rate of appearance of NH_3 is 6.8×10^{-3} g/min, then rate of disappearance of $H_2(g)$ at the same condition will be :

(1) 6.8×10^{-3} gm/ min

(2) 1.02×10^{-2} gm/ min

(3) 6×10^{-4} gm/ min

(4) 1.2×10^{-3} gm/ min

22) Which of the following statements regarding rate constant is correct?

(1) Rate constant always depends on concentration of reactant.

(2) Rate constant is temperature dependent.

(3) For instantaneous reactions, rate constant will be very small.

(4) Rate constant will always depend on pressure or volume of the container

23)

Iodine monochloride reacts with hydrogen according to the equation
$2IC\square(g) + H_2(g) \rightarrow I_2(g) + 2HCl(g)$
The rate data below have been obtained.

[IC□],M	[H_2],M	Initial Rate, Ms^{-1}
0.10	0.10	0.030
0.20	0.10	0.060
0.10	0.050	0.015

What is the rate law for the reaction

(1) Rate=k[ICl] [H_2]

(2) Rate=k[ICl]^2 [H_2]

(3) Rate=k[ICl] [H_2]$^{1/2}$

(4) Rate=k[ICl]^2 [H_2]$^{1/2}$

24) In a zero order reaction half life is 100 sec. After how much time 7/8 fraction of reactant will be reacted?

(1) 300 sec.

(2) 200 sec.

(3) 175 sec.

(4) 25 sec.

25) Consider the Arrhenius equation given below and mark the **CORRECT** option, $k = Ae^{-E_a/RT}$

(1) Rate constant increases exponentially with increasing activation energy and decreasing temperature.

(2) Rate constant remains constant with increasing activation energy and decreasing temperature .

(3) Rate constant increases exponentially with decreasing activation ·energy and decreasing temperature.

(4) Rate constant increases exponentially with decreasing activation energy and increasing temperature.

26) Which of the following is correctly matched?

(1) Basic oxides ⇒ Na_2O, CaO, SO_3

(2) Neutral oxides ⇒ CO_2, NO_2, N_2O

(3) Acidic oxides ⇒ ZnO, SO_2, P_2O_5

(4) Amphoteric oxides ⇒ Al_2O_3, ZnO, SnO_2

27) Among the following species, select the diamagnetic molecule:

(1) B_2

(2) CO

(3) NO

(4) O_2

28) For SO_2, NO_2^- and N_3^-, identify the species in which the central atom is sp-hybridized:

(1) SO_2 only

(2) NO_2^- only

(3) N_3^- only

(4) SO_2 and NO_2^-

29) Given below are two statements:
Statement I: Along a period, the chemical reactivity of elements increases uniformly from group 1

to group 18.
Statement II: Oxides of group 1 elements are generally basic, whereas oxides of many group 17 elements are acidic.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement-I and Statement-II are true.

(2) Statement-I is true but Statement-II is false.

(3) Statement-I is false but Statement-II is true.

(4) Both Statement-I and Statement-II are false.

30) Correct statement will be :
a. Electron gain enthalpy of oxygen will be least negative in their group.
b. N_2O will be example of sub-oxide
c. In the see-saw geometry of SF_4, lone pair occupies equatorial position
d. BF_3 have bond order 1.

(1) a and b only

(2) a, b and c only

(3) b and c only

(4) a, c and d only

31) The maximum ionization energy would be associated with the electronic configuration :

(1) $1s^2, 2s^2, 2p^6, 3s^1$

(2) $1s^2, 2s^2, 2p^5$

(3) $1s^2, 2s^2, 2p^6$

(4) $1s^2, 2s^2, 2p^6, 3s^2$

32) The correct order of increasing metallic character among Si, Be, Mg, Na, P is :

(1) Si < P < Mg < Be < Na

(2) P < Na < Mg < Si < Be

(3) P < Si < Be < Mg < Na

(4) P < Si < Mg < Na < Be

33) In which of the following arrangements the order is not according to the property indicated against it?

(1) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ – Increasing ionic size

(2) B < C < N < O – Increasing first ionization enthalpy

(3) I < Br < F < Cl – Increasing electron gain enthalpy (with negative sign)

(4) Li < Na < K < Rb – increasing metallic radius

34) Arrange the following in increasing order of covalent character - NaCl, $MgCl_2$, $AlCl_3$.

(1) $NaCl < MgCl_2 < AlCl_3$

(2) $MgCl_2 < NaCl < AlCl_3$

(3) $AlCl_3 < MgCl_2 < NaCl$

(4) $NaCl < AlCl_3 < MgCl_2$

35) The radius and charge of each of six ions are shown in the table:

Ion	J^+	L^+	M^{2+}	X^-	Y^-	Z^{2-}
Radius/nm	0.14	0.18	0.15	0.14	0.18	0.15

The ionic solids JX, LY and MZ are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first ?

(1) JX > LY > MZ

(2) JX > MZ > LY

(3) LY > MZ > JX

(4) MZ > JX > LY

36) The corrects increasing order for bond angles among BF_3, PF_3 and ClF_3 is :

(1) $PF_3 < BF_3 < ClF_3$

(2) $BF_3 < PF_3 < ClF_3$

(3) $ClF_3 < PF_3 < BF_3$

(4) $BF_3 = PF_3 < ClF_3$

37) Arrange the noble gases in increasing polarizability : He, Ne, Ar, Xe

(1) He < Ne < Ar < Xe

(2) Ne < He < Ar < Xe

(3) Xe < Ar < Ne < He

(4) Ar < Ne < He < Xe

38) Which molecule is non-polar?

(1) NH_3

(2) H_2O

(3) SO_2

(4) CO_2

39) Which ion is paramagnetic?

(1) Zn^{2+}

(2) Cu^+

(3) Sc^{3+}

(4) Fe^{3+}

40) Which element has smallest atomic radius?

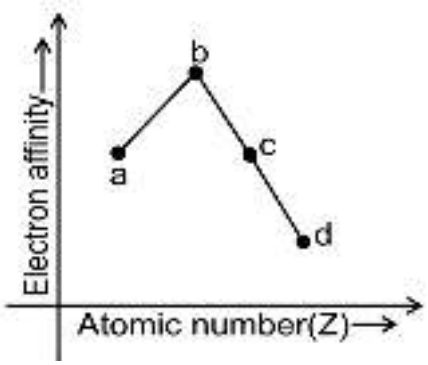
(1) Li

(2) Be

(3) B

(4) F

41)



Where a, b, c, d are elements of group-17 if difference between atomic number of elements d and b is x and difference between atomic number of elements c and a is y. What is the value of x-y.

(1) 10

(2) 9

(3) 11

(4) 12

42) In C_3O_2 molecule z-axis is assumed as molecular axis. The maximum no. π-bonds present in x-z plane.

(1) 1

(2) 2

(3) 3

(4) 4

43) PCl_5 is well known, but NCl_5 is not.
Because

(1) Nitrogen is less reactive than phosphorous

(2) Nitrogen doesn’t have d-orbitals in its valence shell

(3) Catenation tendency is weaker in nitrogen than phosphorous

(4) Size of phosphorous is larger than nitrogen

44) Which compound shows hydrogen bonding?

(1) HCl

(2) HF

(3) HBr

(4) HI

45) Identify the incorrect match:

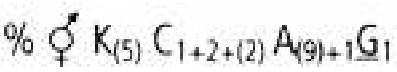
Name	IUPAC Official Name
A. Unnilunium	i. Mendelevium
B. Unniltrium	ii. Livermorium
C. Unnilhexium	iii. Seaborgium
D. Ununbium	iv. Darmstadtium

(1) A, B
(2) B, C
(3) C, D
(4) B, D

BIOLOGY

1) Ashwagandha and Belladonna are medicinal plants belonging to the family :

(1) Liliaceae
(2) Fabaceae
(3) Solanaceae
(4) Brassicaceae

2) The floral formula  represents which family?

(1) Fabaceae
(2) Solanaceae
(3) Liliaceae
(4) Brassicaceae

3) The stamens are united into two bundles (diadelphous) in :

(1) China rose (Malvaceae)
(2) *Citrus*
(3) Pea (Fabaceae)
(4) Mustard (Cruciferae)

4) The region of the root responsible for the growth of the root in length is:

(1) Region of maturation
(2) Root cap
(3) Region of elongation
(4) Region of meristematic activity

5) When stamens are attached to the perianth, as in the flowers of lily, they are called:

(1) Epipetalous

(2) Epiphyllous
(3) Polyandrous
(4) Monoadelphous

6)  Choose the correct option regarding this diagram :-

(1) It is racemose type of inflorescence in which main axis terminates in a flower
(2) It is cymose type of inflorescence in which flowers are borne in a basipetal order
(3) This type of inflorescence shows indefinite growth
(4) This type of inflorescence is commonly seen in all angiospermic plants

7) Which of the following statement is not correct about stem?

(1) Bears node and internode
(2) Can perform vegetative propagation
(3) Develops from the radicle of the embryo
(4) Bears terminal or axillary buds

8) How many have underground stem in the following?
Potato, Ginger, Turmeric, Zaminkand and *Colocasia*

(1) 2
(2) 3
(3) 4
(4) 5

9) Which type of phyllotaxy present in mustard, Guava and *Alstonia* respectively :

(1) Opposite, Alternate and whorled
(2) Alternate, opposite, and whorled
(3) Whorled, opposite and alternate
(4) Alternate, whorled and opposite

10) Axillary bud is derived from :

(1) Root apical meristem
(2) Phellogen
(3) Cork
(4) Shoot apical meristem

11) Leaf mesophyll is included in :-

(1) Epidermal tissue system

(2) Ground tissue system

(3) Vascular tissue system

(4) All of the above

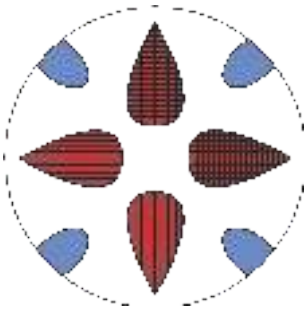
12) Mature and differentiated cells of one of the following contain cytoplasm, but no nucleus?

(1) Xylem parenchyma

(2) Companion cells

(3) Xylem vessels

(4) Sieve tubes



13) Which type of primary xylem is present in given figure :-

(1) Exarch

(2) Endarch

(3) Mesarch

(4) Concentric

14)

In monocotyledon plant vascular bundles are closed type because :-

(1) Vascular bundles have cambium and cambium possess the ability to form secondary xylem and phloem

(2) Vascular bundles have no cambium and they do not form secondary xylem and phloem

(3) Vascular bundles have cambium and cambium do not possess the ability to form secondary tissues

(4) Both (1) and (2)

15) Which of the following option gives the **correct** categorisation of the tissue systems found in plants :-

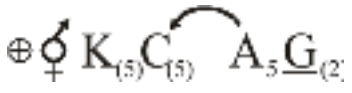
	A Epidermal tissue system	**B** Ground tissue system	**C** Vascular tissue system
(1)	Bulliform cells	Primary xylem	Cambium
(2)	Stomata	Pericycle	Medullary Rays
(3)	Trichomes	Hypodermis	Vascular Bundles
(4)	Root hair	Endodermis	Epiblema

(1) 1

(2) 2

(3) 3

(4) 4

16) The floral formula  is that of :-

(1) Tobacco

(2) Tulip

(3) Soyabean

(4) Sunhemp

17)

Pulses yielding main family of plants is :-

(1) Poaceae (Graminae)

(2) Cucurbitaceae

(3) Liliaceae

(4) Papilionaceae (Fabaceae)

18) Match the column-I with column II :

Column-I		**Column-II**	
(P)	Tetradynamous	(a)	Leguminosae
(Q)	Diadelphous	(b)	Malvaceae
(R)	Monoadelphous	(c)	Cruciferae
(S)	Syngenesious	(d)	Asteraceae

(1) P→c, Q→a, R→b, S→d

(2) P→a, Q→c, R→b, S→d

(3) P→c, Q→a, R→d, S→b

(4) P→b, Q→d, R→a, S→c

19) (A) Forms the major component within organs.
(B) Cells are generally isodiametric.
(C) Cell walls are thin and made up of cellulose.
(D) Lignin is always absent.
(E) Provide mechanical support to the growing aerial parts of the monocot plant.
(F) They are usually with protoplast and dead.
How many statements from above are correct for sclerenchyma tissue ?

(1) One

(2) Two

(3) Three

(4) Four

20) Select correct statements :-
(i) Growth in plants is largely restricted to specialized regions.
(ii) Meristem which occurs between mature tissues known as apical meristem.
(iii) Ground tissue of leaves is known as mesophyll.
(iv) Dicot roots show secondary growth.
(v) Size of vascular bundles is highly variable in monocot leaves.

(1) i, ii, iii, iv
(2) i, iii, iv, v
(3) i, iii, iv
(4) ii, iii, v

21) Which of the following taxonomic categories are common for *Triticum* and *Mangifera* ?

(1) Division, Kingdom
(2) Division, Class
(3) Order, Class
(4) Family, Orders

22) Choose correct option for A, B and C ?
Solanum + A = Solanaceae
Panthera + B = C

	A	**B**	**C**
(1)	*Petunia*	*Canis*	Canidae
(2)	*Datura*	*Felis*	Felidae
(3)	*Mangifera*	*Canis*	Canidae
(4)	*Oryza*	*Felis*	Felidae

(1) 1
(2) 2
(3) 3
(4) 4

23) Consciousness is considered a defining property of living organisms because :

(1) Except plants, all eukaryotes can sense their surroundings
(2) Most complex organisms are unable to respond environmental stimuli
(3) All organisms are aware of their surroundings and respond to external environmental stimuli
(4) Except microbes, all organisms show consciousness

24) Lichens are the best indicators of :-

(1) Water pollution
(2) Soil pollution
(3) Air pollution

(4) Noise pollution

25) Biological concept of species is followed by :-

(1) *Homo sapiens*
(2) *Acetobacter*
(3) Mule
(4) *Mycoplasma*

26) Find the incorrect statement from the following -

(1) Aristotle was the earliest to attempt a more scientific basis for classification.
(2) Fungi have chitin in their wall while green plants have cellulose.
(3) Dinoflagellates float passively in water while diatoms have two flagella for floating.
(4) In euglenoids, both photosynthetic and heterotrophic nutrition is present

27) There is a table given below according to Whittaker's five kingdom system. What are (A) and (B) in the given table :

Character	**Monera**	**Protista**	**Fungi**
Body organisation	Cellular	Cellular	**(A)**
Cell wall	Present	**(B)**	Present

Option :

	(A)	**(B)**
(1)	Unicellular or multicellular	Absent
(2)	Multicellular/ loose tissue	Present in some members
(3)	Multicellular/ loose tissue	Present in all members
(4)	only unicellular	Absent in all members

(1) 1
(2) 2
(3) 3
(4) 4

28) Main criteria used by R.H. Whittaker in his five kingdom classification was not the :-

(1) Cell structure
(2) Body organization
(3) Type of spore formation
(4) Mode of nutrition

29) Select the incorrect statement with respect to diatoms.

(1) They float passively in water currents

(2) Mostly members are photosynthetic

(3) They have flagella for movement

(4) They are found in marine environments

30) Select true regarding thermoacidophiles-
(a) It is a cell wall less eubacterium.
(b) They are members of archaea.
(c) They are found in certain hydrothermal springs.
(d) They inhabit alkaline hot springs.
(e) Their enzyme activate at acidic pH.

(1) a, c and d

(2) b, c and e

(3) a, b and c

(4) b, c and d

31) Consider the following statements about *Mycoplasma* :-
(i) It is pleomorphic organism which lacks cell wall.
(ii) *Mycoplasma* is the smallest living organism.
(iii) They can survive without oxygen.
(iv) Many *Mycoplasma* are pathogenic in animals and plant.
Choose the **correct** statements :-

(1) I, II & III

(2) II, III and IV

(3) I, II & IV

(4) I, II, III and IV

32) In five kingdom system of classification blue green algae are included in :-

(1) Animalia

(2) Plantae

(3) Monera

(4) Protista

33) Members of which group are related with basidiomycetes ?

(1) Mushroom, *Ustilago, Aspergillus*

(2) *Puccinia*, *Agaricus, Aspergillus*

(3) *Puccinia*, *Ustilago, Agaricus*

(4) *Ustilago, Aspergillus, Agaricus*

34) Which one of the following diagram represents the slime mould :-

(1) 

(2) 

(3) 

(4) 

35) Choose the incorrect statement of following :-

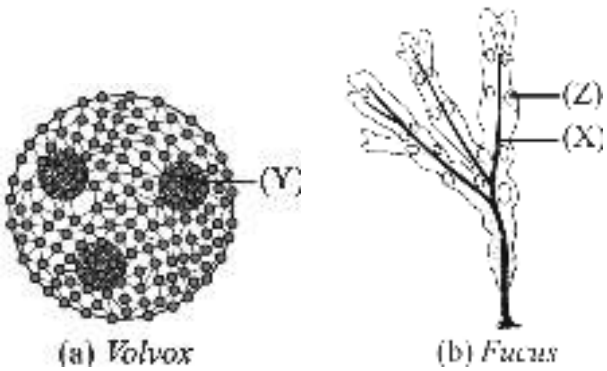
(1) *Chlorella* and *Spirulina* are protein rich unicellular members.

(2) Mosses reduce the impact of falling rain and prevent soil erosion.

(3) In homosporous pteridophytes, gametophytic generation is highly reduced, mostly heterotrophic and is confined to only a limited number of cells.

(4) In gymnosperm the nucellus is protected by envelopes and the composite structure is called an ovule.

36)



Identify the (X), (Y) and (Z) in these figures :-

(1) X–DNA, Y–Sex organ, Z-Gemma cup

(2) X–Mid rib, Y–Daughter colony, Z-Air bladder

(3) X–DNA, Y–Oogonium, Z-Frond

(4) X–Air bladder, Y–Archegonium, Z-Stipe

37) Which of the following statement is correct ?

(1) In *Sphagnum*, generally asexual reproduction takes place by green, multicellular buds called gemmae.

(2) The sporophyte of bryophytes is differentiated into foot, seta and capsule and it is a free-living generation

(3) Primary protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore.

(4) Liverworts have an elaborate mechanism of seed dispersal.

38) Which of the following feature(s) is/are exclusive in angiosperms :-
(A) Seed
(B) Double fertilization
(C) Female gametophyte
(D) Presence of ovule
(E) Formation of endosperm
(F) Presence of companion cells in phloem
Tick the right one ;-

(1) A, B, C, E only

(2) B, C, D, E only

(3) B & F only

(4) A, B, C, D only

39) In the life cycle of mosses, the gametophyte has two stages (A and B). These stages can be called:

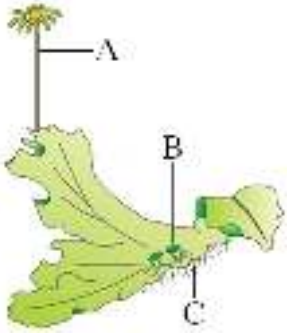
(1) A-Protonema; B-Leafy stage

(2) A-Protonema; B-Sporogonium

(3) A-Sporophyte; B-Gametophyte

(4) A-Zygote; B-Spore mother cell

40) Identify labelling A, B and C in figure given below and choose the correct option :-



	A	B	C
(1)	Antheridiophore	Gemma cup	Rhizoids
(2)	Archegoniophore	Gemma cup	Roots
(3)	Antheridiophore	Rhizoids	Gemma cup

(4)	Archegoniophore	Gemma cup	Rhizoids

(1) 1

(2) 2

(3) 3

(4) 4

41) Correct sequence of spermatogenesis is:

(1) Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa

(2) Primary spermatocyte → Spermatogonia → Spermatid → Sperm

(3) Spermatid → Spermatogonia → Spermatozoa

(4) Secondary spermatocyte → Spermatogonia → Sperm

42) 1. Menstrual cycle lasts about 28 days.
2. Ovulation occurs on 14th day.
3. Estrogen level rises before ovulation.
4. Progesterone dominates after ovulation.
5. Menstruation occurs if fertilization takes place.
Choose the correct statements :

(1) 1, 3 and 5

(2) 1, 2, 3 and 4

(3) 2, 4 and 5

(4) All of these

43) **Assertion (A):** Oogenesis starts before birth.
Reason (R): Primary oocytes are formed during fetal life.

(1) Both true, R explains A

(2) Both true, R does not explain A

(3) A true, R false

(4) A false, R true

44) 1. Testes produce sperms and testosterone.
2. Epididymis stores sperms.
3. Vas deferens transports sperms.
4. Prostate gland produces ovum.
Choose the correct statements :

(1) 1, 3, 4

(2) 1, 2, 3

(3) 2, 3, 4

(4) All of these

45) 1. Ovary produces ova.
2. Fallopian tube is the site of fertilization.

3. Uterus supports embryo development.
4. Uterus opens into vagina through wide cervix.
Choose the correct statements :

(1) 1, 3, 4
(2) 2, 3, 4
(3) 1, 2, 3
(4) All of these

46) **Statement A:** Fertilization occurs in ampulla.
Statement B: Implantation occurs in uterus.

(1) Both statement A and statement B are incorrect.
(2) Both statement A and statement B are correct.
(3) Statement A is correct, statement B is incorrect.
(4) Statement A is incorrect, statement B is correct.

47) Match the columns -

Column I	Column II
A. 1st month	1. Heart formed
B. 2nd month	2. Limbs develop
C. 3rd month	3. Major organs formed

(1) A-1, B-3, C-2
(2) A-2, B-1, C-3
(3) A-3, B-2, C-1
(4) A-1, B-2, C-3

48) Parturition is initiated by:

(1) Thyroxine
(2) Insulin
(3) Oxytocin hormone
(4) Estrogen

49) A mother produces milk after childbirth. Which hormone is responsible?

(1) Prolactin
(2) Oxytocin
(3) Estrogen
(4) Progesterone

50) Match the columns w.r.t. structure of sperm and its function.

Column I	Column II

A. Head	1. Genetic material
B. Acrosome	2. Enzymes
C. Tail	3. Motility

(1) A-2, B-1, C-3

(2) A-3, B-2, C-1

(3) A-1, B-2, C-3

(4) A-1, B-3, C-2

51) 1. Reproductive health includes physical and social well-being.
2. Awareness programs help in population control.
3. Sexually transmitted diseases are part of reproductive health issues.
4. Reproductive health only focuses on pregnancy.
5. Government promotes family planning programs.
Choose the correct statements -

(1) 1, 2, 3, 5

(2) 1, 3, 4

(3) 2, 4, 5

(4) All of these

52) 1. Rhythm method avoids intercourse during fertile period.
2. Lactational amenorrhea works after childbirth.
3. Coitus interruptus is reliable 100%.
4. Natural methods involve no devices.
Choose the correct option:

(1) 1, 3, 4

(2) 1, 2, 4

(3) 2, 3, 4

(4) All of these

53) Match the columns -

Column I	Column II
A. Condom	1. Prevents STI
B. Diaphragm	2. Blocks sperm entry
C. IUDs	3. Suppress sperm motility

(1) A-2, B-1, C-3

(2) A-3, B-2, C-1

(3) A-1, B-3, C-2

(4) A-1, B-2, C-3

54) Which is correct difference between vasectomy and tubectomy?

(1) Vasectomy in males; Tubectomy in females

(2) Both in males

(3) Both in females

(4) Vasectomy blocks eggs

55) 1. MTP is legal in India under conditions.
2. It helps in unwanted pregnancy control.
3. It is always safe without risk.
4. It should be done by medical professionals.
Choose the correct option:

(1) 1, 3, 4

(2) 1, 2, 4

(3) 2, 3, 4

(4) All of these

56) **Statement I:** STIs spread through sexual contact.
Statement II: STIs cannot be prevented.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

57) Match the columns -

Column I	**Column II**
A. IVF	1. Fertilization outside body
B. ZIFT	2. Zygote transfer
C. GIFT	3. Gamete transfer

(1) A-3, B-2, C-1

(2) A-2, B-1, C-3

(3) A-1, B-2, C-3

(4) A-1, B-3, C-2

58) 1. Copper-T is IUD.
2. IUD prevents fertilization.
3. IUD is hormonal only.
Choose the correct statements -

(1) 1, 2 only

(2) 2, 3 only

(3) 1, 3 only

(4) All of these

59) Small doses of progesterone pills are used for:

(1) Contraception
(2) Digestion
(3) Respiration
(4) Growth

60) **Assertion (A):** STIs can be prevented by safe sex practices.
Reason (R): Use of condoms reduces transmission risk.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true and reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both Assertion and reason are false.

61) Match the columns -

Column I	Column II
A. Porifera	1. Pores all over body
B. Cnidaria	2. Radial symmetry
C. Arthropoda	3. Jointed appendages
D. Echinodermata	4. Water vascular system

(1) A-3, B-2, C-1, D-4
(2) A-2, B-1, C-4, D-3
(3) A-1, B-2, C-3, D-4
(4) A-1, B-3, C-2, D-4

62) Match the columns -

Column I	Column II
A. Pisces	1. Gills
B. Amphibia	2. Dual habitat
C. Reptilia	3. Dry scaly skin
D. Aves	4. Feathers

(1) A-1, B-2, C-3, D-4
(2) A-2, B-1, C-4, D-3
(3) A-3, B-2, C-1, D-4
(4) A-1, B-3, C-2, D-4

63) Match the columns -

Column I	Column II

A. Asymmetry	1. Sponges
B. Radial symmetry	2. Jellyfish
C. Bilateral symmetry	3. Humans

(1) A-2, B-1, C-3

(2) A-3, B-2, C-1

(3) A-1, B-2, C-3

(4) A-1, B-3, C-2

64) 1. Notochord is absent.
2. They may have open circulatory system in all.
3. Segmentation is present in all.
4. Respiration varies in groups.
5. Skeleton may be external or absent.
Choose the correct statements regarding non-chordates :

(1) 1, 4, 5

(2) 1, 3, 4

(3) 2, 3, 5

(4) All of these

65) 1. Notochord present at least in embryonic stage.
2. Dorsal hollow nerve cord present.
3. Pharyngeal gill slits present.
4. Ventral nerve cord present.
5. Post-anal tail present.
Choose the correct statements w.r.t. chordates :

(1) 1, 2, 3, 5

(2) 1, 3, 4

(3) 2, 4, 5

(4) All of these

66) **Assertion (A):** Annelids and arthropods are coelomates.
Reason (R): Body cavity is lined by mesoderm.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both Assertion and reason are false.

67) **Assertion (A):** Porifera are called pore-bearing animals.
Reason (R): Body has many pores called ostia.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both Assertion and reason are false.

68) Which belong to Arthropoda?

(1) Starfish, Sea urchin
(2) Earthworm, Leech
(3) Insects, Crustaceans
(4) Sponge, Hydra

69) Which are mammals?

(1) Frog, Snake
(2) Whale, Bat
(3) Crow, Eagle
(4) Fish, Shark

70) **Statement I:** Echinoderms are marine animals.
Statement II: They show bilateral symmetry in adults.

(1) Both statement I and statement II are correct.
(2) Statement I is incorrect but statement II is correct.
(3) Statement I is correct but statement II is incorrect.
(4) Both statement I and statement II are incorrect.

71) Match the columns (frog) :

Column I	Column II
A. Skin	1. Respiration
B. Liver	2. Bile secretion
C. Heart	3. Circulation
D. Kidney	4. Excretion

(1) A-4, B-2, C-3, D-1
(2) A-1, B-3, C-2, D-4
(3) A-2, B-1, C-3, D-4
(4) A-1, B-2, C-3, D-4

72) Match the columns (Cockroach) :

Column I	Column II
A. Trachea	1. Respiration
B. Malpighian tubules	2. Excretion
C. Hemolymph	3. Circulation

(1) A-2, B-1, C-3

(2) A-1, B-2, C-3

(3) A-3, B-2, C-1

(4) A-1, B-3, C-2

73) Match the columns -

Column I	Column II
A. Frog	1. Amphibian
B. Cockroach	2. Arthropod
C. Snake	3. Reptilian
D. *Echinus*	4. Echinodermata

(1) A-1, B-2, C-3, D-4

(2) A-2, B-1, C-4, D-3

(3) A-1, B-3, C-2, D-4

(4) A-3, B-2, C-1, D-4

74) 1. Frog is cold-blooded.
2. Skin is glandular.
3. Respiration only by lungs.
4. Heart is three-chambered.
Choose the correct option w.r.t. frog :

(1) 1, 3, 4

(2) 2, 3, 4

(3) 1, 2, 4

(4) All of these

75) 1. Body divided into head, thorax, abdomen.
2. Open circulatory system present.
3. Respiration by gills.
4. Excretion by Malpighian tubules.
Choose the correct option w.r.t. Cockroaches :-

(1) 1, 3, 4

(2) 1, 2, 4

(3) 2, 3, 4

(4) All of these

76) 1. Frog undergoes metamorphosis.
2. Tadpole breathes through gills.
3. Adult frog is ammonotelic.
4. Frog excretes urea.
Choose the correct option w.r.t. frog :

(1) 1, 3, 4

(2) 2, 3, 4

(3) 1, 2, 4

(4) All of these

77) Frog belongs to :

(1) Amphibia

(2) Reptilia

(3) Aves

(4) Mammalia

78) In Cockroach respiration occurs through:

(1) Gills

(2) Trachea

(3) Lungs

(4) Skin

79) Frog heart has :

(1) 2 chambers

(2) 3 chambers

(3) 4 chambers

(4) 1 chamber

80) **Assertion (A):** Cockroach has open circulatory system.
Reason (R): Blood flows freely in body cavity.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both Assertion and reason are false.

81) **Assertion (A):** Double fertilization is unique to angiosperms.
Reason (R): Only in angiosperms one male gamete fuses with egg and the other with secondary nucleus.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not correct explanation.

(3) A is true but R is false.

(4) A is false but R is true.

82) **Assertion (A):** Synergids help in pollen tube entry into embryo sac.
Reason (R): Filiform apparatus guides the pollen tube.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not correct explanation.

(3) A is true but R is false.

(4) A is false but R is true.

83) Which of the following statements are correct?
a. Tapetum provides nourishment to developing pollen grains.
b. Endothecium helps in dehiscence of anther.
c. Middle layers persist in mature anther.
d. Microspore tetrads are formed after meiosis.

(1) a, b and d only

(2) a and d only

(3) b and c only

(4) All of these

84)

Identify incorrect statements:
a. Egg apparatus consists of one egg cell and two synergids.
b. Antipodals are present near micropyle.
c. Central cell contains two polar nuclei.
d. Embryo sac is generally 8-nucleate and 7-celled.

(1) b only

(2) a and b

(3) c only

(4) None of these

85) Select correct statements regarding pollination:
a. Xenogamy brings genetically different pollen grains.
b. Geitonogamy is genetically similar to autogamy.
c. Autogamy requires external pollinating agents.
d. Xenogamy increases variation.

(1) a, b and d

(2) a and c

(3) b and c

(4) All of these

86) Choose the correct statements:
a. Apomixis produces seeds without fertilization.
b. Polyembryony means many embryos in one seed.
c. Adventive embryony develops from nucellus.
d. Hybrid varieties can not retain traits generation after generation if they are producing apomictic seeds.

(1) a, b and c only

(2) b and d only

(3) a and c only

(4) All of these

87)

Read the statements:
a. Persistent nucellus in seed is called perisperm.
b. Coconut water is free nuclear endosperm.
c. Albuminous seeds lack endosperm.
d. Aleurone layer is protein rich.
Choose the correct option:

(1) a, b and d only

(2) a and c only

(3) b and d only

(4) All of these

88) Match Column I with Column II.

Column I	Column II
A. Micropyle	i. Pollen tube's entry
B. Chalaza	ii. Basal part of ovule
C. Funicle	iii. Attachment stalk
D. Hilum	iv. Scar on seed

(1) A - ii, B - iii, C - iv, D - i

(2) A - i, B - ii, C - iv, D - iii

(3) A - i, B - ii, C - iii, D - iv

(4) A - ii, B - i, C - iii, D - iv

89) Match the column I with column II and select the correct option.

Column I	Column II
A. *Hydrilla*	i. Insect pollination
B. *Amorphophallus*	ii. Water pollination
C. Coleoptile	iii. Protects plumule
D. Coleorhiza	iv. Protects radicle

(1) A - ii, B - iii, C - iv, D - i

(2) A - iii, B - iv, C - i, D - ii

(3) A - iv, B - iii, C - ii, D - i

(4) A - ii, B - i, C - iii, D - iv

90) Which plant does not show cleistogamy?

(1) *Viola*

(2) *Commelina*

(3) Sunhemp

(4) *Oxalis*

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	4	1	1	3	1	2	4	4	1	1	4	1	1	3	1	2	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	2	2	1	4	4	2	2	1	3	3	1	1	3	2	3	2	3	3
Q.	41	42	43	44	45															
A.	2	4	4	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	4	2	1	2	2	1	3	1	2	3	1	2	4	2	4	1	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	1	3	4	4	2	3	3	2	3	3	2	1	4	3	1	4	4	4
Q.	86	87	88	89	90															
A.	1	2	2	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	3	2	2	3	4	2	4	2	4	1	2	3	1	4	1	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	3	3	1	3	2	3	3	2	4	3	3	1	3	2	3	3	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	1	2	3	2	4	3	1	3	1	2	4	1	2	3	3	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	3	1	1	1	1	3	2	3	4	2	1	3	2	3	1	2	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	1	1	1	1	1	3	4	3										

SOLUTIONS

PHYSICS

1) **Explaining :** For system equilibrium net force on each particle must be zero.
Concept : This question is based on system equilibrium
Solution :

$$\vec{F}_Q = 0$$
$$\vec{F}_{QQ} + \vec{F}_{QQ} + \vec{F}_{Qq} = 0$$
$$\sqrt{3}\frac{kQQ}{a^2} + \frac{kQq}{\left(\frac{a}{\sqrt{3}}\right)^2} = 0$$
$$\sqrt{3}Q = -3q$$
$$\frac{q}{Q} = -\frac{1}{\sqrt{3}}$$

Final Answer : (2)

2) **Explaining :** find new force of repulsion
Concept : This question is based on coulomb's law
Solution :

$$\frac{kq_1q_2}{r^2} = 100$$

After q_1 increase by 10% & q_2 decreased by 10%

$$F = \frac{k(1.1q_1)(0.9q_2)}{r^2}$$
$$F = \frac{kq_1q_2(1.1 \times 0.9)}{r^2}$$
$$F = 100 \times (0.99) = 99 \text{ N}$$

Final Answer : (3)

3) **Explaining :** The electric field in a region shown here is given by $\vec{E} = E_0 x^{\frac{3}{2}}\hat{i}$ volt/m. Find total electric flux through the cube
Concept : This question is based on flux
Solution :

$\phi_{net} = \phi_{in} + \phi_{out}$
$\phi_{in} = 0$ (at x = 0, E = 0)
$\phi_{out} = E_0 (a)^{3/2} (a^2)$ (at x = a, $E = E_0 a^{3/2}$)
$E_0 a^{7/2} \quad \Rightarrow \phi = E_0 a^{7/2}$

Final Answer : (4)

4) **Explaining :** surface charge density $\sigma = \sigma_0 r$, Find total charge on the disc
Concept : This question is based on surface charge density
Solution :

$$Q = \int \sigma dA$$

$$Q = \int_0^R \sigma_0 r 2\pi r dr$$

$$Q = 2\pi\sigma_0 \int_0^R r^2 dr$$

$$Q = \frac{2\pi\sigma_0 R^3}{3}$$

Final Answer : (1)

5) **Question Explain:** find flux through cylinder

A straight wire with linear charge density λ = 8.85μC/m passes through a cylindrical surface of height 4 m . We're asked to compute the electric flux through the cylinder.

Concept Based:

$$\phi = \frac{q_{en}}{\epsilon_0}$$

Gauss's law relates electric flux through a closed surface to the net charge enclosed by the surface. The geometry only matters for calculating enclosed charge.

Formula Used:

$\Phi = \frac{q_{enclosed}}{\varepsilon_0}$, where $q_{enclosed} = \lambda.L$

Calculation:

Given: λ = 8.85 μC/m = 8.85 × 10^{-6} C/m

Length L = 5m (since the wire enters at a slant)

$$L = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5m$$

$\varepsilon_0 = 8.85 \times 10^{-12} C^2/N.m^2$

Now,

$$q_{enclosed} == \lambda.L = (8.85 \times 10^{-6}).5 = 44.25 \times 10^{-6} C$$

$$\Phi = \frac{44.25 \times 10^{-6}}{8.85 \times 10^{-12}} = 5 \times 10^6 volt/m$$

Final Answer: Option (1) is correct

5 × 10^6 volt-m

6) **Explaining :** The electric field is 100 V/m, at a distance of 20 cm from the centre of a dielectric sphere of radius 10 cm. Then E at 3 cm distance from the centre of sphere

Concept : This question is based on

$$E_{in} = \frac{KQr}{R^3}$$

$$E_{wt} = \frac{KQ}{r^2}$$

Solution :

At r = 20 cm $\Rightarrow E = \frac{kQ}{r^2} = 100V/m \Rightarrow kQ = 100r^2$

At r' = 3 cm $\Rightarrow E = \frac{kQr}{R^3} = \frac{(100r^2)r'}{R^3}$

$$= \frac{100 \times (20 \times 10^{-2})^2 \times 3 \times 10^{-2}}{(10 \times 10^{-2})^3} = 120V/m$$

Final Answer : (3)

7) **Explaining :** To find the electric field from given potential v/s position eq^{n}.

Concept : This question is based on $\vec{E} = -\vec{\nabla}(V)$.

Solution : $E_x = \frac{-\partial v}{dx} = -4xk \Rightarrow E_x = -4 \times 1 = -4k$

$E_y = \frac{-\partial v}{dy} = +4yk \Rightarrow E_y = 4 \times 1 = 4k$

$E_z = \frac{-\partial v}{dz} = -2zk \Rightarrow E_z = -2 \times 1 = -2k$

$E = \left(\sqrt{4^2 + 4^2 + 2^2}\right) k = 6k$

Final Answer : (1)

8) **Explaining :** To find relation b/w potential difference & distance as given diagram.

Concept : This question is based on $E = -\frac{dV}{dr}$.

Solution : $\because V_1 - V_2 = V_2 - V_3$ (given)

As we go along positive x-axis, electric field increases, E.F.L. becomes denser so distance so distance between equal potential surfaces.

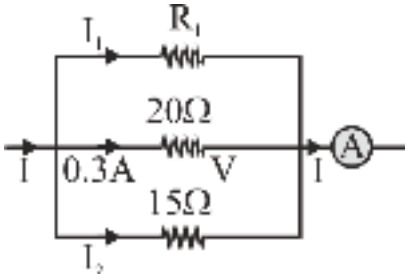
Decreases is $x_1 > x_2$.

Final Answer : (2)

9) **Explaining :** In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammeter reads 0.8 ampere. What is the value of R_1 ?

Concept : This question is based on Ohms law.

Solution : $V = 20 \times 0.3 = 6V$



$I_2 = \frac{6}{15} = 0.4$ A

$I = I_1 + 0.3 + I_2$

$\Rightarrow 0.8 = I_1 + 0.3 + 0.4$

$\Rightarrow I_1 = 0.1$ A; $R_1 = \frac{V}{I_1} = \frac{6}{0.1} = 60\Omega$

Final Answer : (4)

10) **Explaining :** For given circuit find $V_A - V_B$:-

Concept : This question is based on KVL.

Solution : $V_A - 2 \times 6 - 12 - 9 \times 2 + 4 - 10 = V_B$

$\Rightarrow V_A - V_B = 48V$

Final Answer : (4)

11) **Explaining :** Power dissipated in the circuit shown in the figure is 67.5 watt .The value of

R is :-

Concept : This question is based on $P = \frac{V^2}{R}$.

Solution : $P_T = P_1 + P_2$

$67.5 = \frac{(15)^2}{10} + \frac{(15)^2}{R} \Rightarrow R = 5\Omega$

Final Answer : (1)

12) **Explaining :** Two sources of emf 6 V and internal resistance 3 Ω and 2 Ω are connected to an external resistance R as shown. If potential difference across battery A is zero, then value of R is :

Concept : This question is based on $i = \frac{E_{net}}{R_{net}}$.

Solution : $i = \frac{E_{net}}{R_{net}} = \frac{6+6}{3+2+R} = \frac{12}{5+R}$

$(V)_A = E_A - ir_A = 0$

$E_A = ir_A$

$6 = \left(\frac{12}{5+R}\right)(3) \Rightarrow 12 = 10 + 2R$

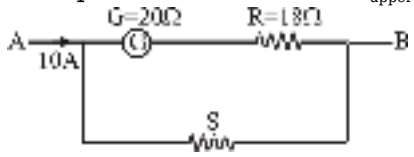
$R = 1\Omega$

Final Answer : (1)

13) **Explaining :** Full scale deflection current for galvanometer is 1 Amp. When shunt 'S' is connected then, what should be the value of shunt resistance so that galvanometer shows half scale deflection.

Concept : This question is based on $V_{upper} = V_{lower\ part}$

Solution :



$i_g = 0.5A$

$0.5(20 + 18) = (10 - 0.5)s$

$19 = 9.5s$

$\Rightarrow$ $S = 2\Omega$

Final Answer : (4)

14) **Explaining :** Circuit diagram of metre bridge is shown in figure. The null point is found at a distance of 40 cm from A. If now a resistance of 12Ω is connected in parallel with S, the null point occurs at 64 cm. Determine the value of R and S.

Concept : This question is based on metre bridge.

Solution :

$\frac{40}{60} = \frac{R}{S}$, $\frac{2}{3} = \frac{R}{S}$ .....(1)

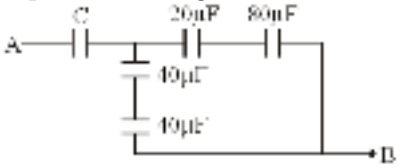
$\frac{64}{36} = \frac{R(12+S)}{12S}$ $\frac{16}{9} = \frac{R(12+S)}{12S}$ ....(2)

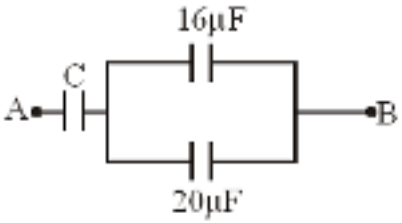
(1)/(2)

$S = 20\Omega$, $R = \frac{40}{3}\Omega$

Final Answer : (1)

15) **Explanation:** Find the value of an unknown capacitor C in a network where the effective capacitance between A and B is 12μF.
Concept: Solving capacitor networks using series and parallel combination rules.
Solution:
Equivalent diagram





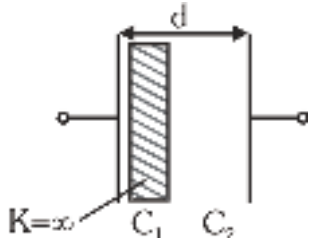
≡

$$\equiv \frac{C \times 36}{C + 36} = 12$$

C = 18 μF
Final Answer : (1)

16) **Explanation:** Find the ratio of new to old capacitance (C'/C) when a copper slab of thickness d/3 is inserted into a parallel plate capacitor.
Concept: A conductor acts as a dielectric with K = ∞ . The new capacitance is given by C' $= \frac{A\epsilon_0}{d - t + t/K}$.
Solution :

$$C = \frac{A\varepsilon_0}{d}$$



$$C' = \frac{A\varepsilon_0}{d - t + \frac{t}{K}}$$

$K_{metal} = \infty$

$$C' = \frac{A\varepsilon_0}{d - \frac{d}{3}} = \frac{3}{2}C$$

Final Answer : (3)

17) **Question Explanation:**
Redistribution of charge between capacitors when connected together.

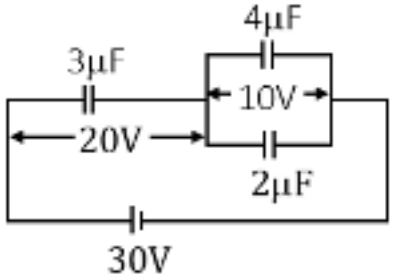
Concept:
Common Potential and Charge Sharing.

Solution:
First, capacitor A is fully charged by the source.
When connected to B, charge flows until both reach the same potential.
Total charged remains conserved.
Calculation:

$Q_{total} = C_A V = 3\ \mu F \times 10\ V = 30\ \mu C$ $V_{common} = \frac{Q_{net}}{C_{net}} = \frac{30}{3+2} = 6V$
$Q_A = 3 \times 6 = 18\ \mu C$; $Q_B = 2 \times 6 = 12\ \mu C$

Final answer: option(1)

18) **Explanation:** For the circuit shown, match the charge supplied by the battery and the charges on specific capacitors.
Concept: Capacitors in series have the same charge, and capacitors in parallel have the same voltage. The total charge is $Q = C_{eff} V$.
Solution:



(i) $Q_{battery} = C_{eff} V$
$= 2 \times 30 = 60\ \mu C$
(ii) $Q_{4\mu F} = 4 \times 10$
$= 40\ \mu C$
(iii) $Q_{2\mu F} = 2 \times 10$
$= 20\ \mu C$
(iv) $Q_{3\mu F} = 3 \times 20$
$= 60\ \mu C$
Final Answer : (2)

19) **Explaining :** Bulb with lower power rating dissipates more power in series.
Concept : This question is based on $P = I^2R$
Solution :

$R_1 = \frac{200 \times 200}{25} = 160\,\Omega$
$R_2 = \frac{200 \times 200}{100} = 400\,\Omega$
$R_{eff} = 2000\Omega$
$i = \frac{400}{2000} = \frac{1}{5}A$
$P_1 = \frac{1}{25} \times 1600 = 64W > 25W$
$P_2 = \frac{1}{25} \times 400 = 16W < 100$
 25W bulb will fuse.

Final Answer : (1)

20)

Explaining : After disconnecting the battery, charge becomes constant. Increasing distance reduces capacitance and increases voltage, but electric field stays constant.
Concept : This question is based on

A. Battery disconnected ⇒ Q = constant
B. $C = \varepsilon_0 A/d, V = Q/C$
C. $E = \sigma/\varepsilon_0 = Q/A\varepsilon_0 \Rightarrow$ constant

Solution :
Electric field $E = V/d = Q/(\varepsilon_0 A)$ remains unchanged.
Final Answer : (4)

21) **Explaining :** Electric fields due to each plate are in opposite directions. Net field:
$E = \frac{Q_1 - Q_2}{2\varepsilon_0 A}$
Then V = Ed = $\frac{Q_1 - Q_2}{2C}$
Concept : This question is based on $E = \frac{Q_1 - Q_2}{2\varepsilon_0 A}$
$V = Ed = \frac{Q_1 - Q_2}{2C}$ using $C = \varepsilon_0 A/d$
Solution :
Within the plates electric fields due to charges Q_1 and Q_2 are
$E_1 = \frac{Q_1}{2\varepsilon_0 A}$ and $E_2 = \frac{Q_2}{2\varepsilon_0 A}$
As these fields are in opposite directions and $Q_1 > Q_2$, the net electric field within the plates is
$E = E_1 - E_2 = \frac{1}{2\varepsilon_0 A}(Q_1 - Q_2)$
Hence $V = Ed = \frac{d}{2\varepsilon_0 A}(Q_1 - Q_2) = \frac{Q_1 - Q_2}{2C}$
$\left(\because C = \frac{\varepsilon_0 A}{d}\right)$
Final Answer : (4)

22) **Explaining :** A parallel plate capacitor of value 1.77 μF is to be designed using a dielectric material of dielectric constant = 200, break through strength = 3×10^6 Vm^{-1}. In order to make such a capacitor, which can withstand a potential difference of 20V across the plates,
Concept : This question is based on parallel plate capacitor
Solution :
$C = \left(\frac{\varepsilon_0 \varepsilon_r A}{d}\right)$
C = 1.77μF & V = 20 V
⇒ & E × d = V

$$C = \frac{\varepsilon_0 \varepsilon_r A}{\left(\frac{V}{E}\right)}$$

$\Rightarrow$

$$A = \frac{CV}{\varepsilon_0 \varepsilon_r E} = 2 \times 10^{-2}$$

$$d = \frac{\varepsilon_0 \varepsilon_r A}{C} = 10^{-5}$$

Final Answer : (2)

23) **Explaining :** Find acceleration and velocity.

Concept : $v = \frac{dx}{dt}, a = \frac{dv}{dt}$

Solution : $v = \frac{dx}{dt} = 12t^2 - 6t$

and $a = \frac{dv}{dt} = 24t - 6$

 v at t = 2s is $12(2)^2 - 6(2)$ i.e., 36 ms^{-1}

and a at t = 2s is $24 \times 2 - 6$ i. e., 42 m/s^2

Final Answer : (2)

24) **Explaining :** Find $\frac{x}{y}$ where x = maximum height and y = time of flight.

Concept : $S = ut + \frac{1}{2}at^2$

and $t_A + t_D$ = Time of flight

Solution : $T = t_1 + t_2 \Rightarrow T = 12$ sec

$$H_{max} = \frac{1}{2}g\left(\frac{12}{2}\right)^2 = \frac{1}{2}(10)(6)^2 = 180m$$

$$x/y = \frac{180}{12} = 15$$

Final Answer : (2)

25) **Explaining :** Find velocities of trains.

Concept : When $\vec{a}_{rel} = \vec{0}$ then $\boxed{v_{rel} = \frac{S_{rel}}{t}}$.

Solution :

$3v_A = v_B$, $S_{rel} = v_{rel}\, t \Rightarrow 100 + 60 = (v_A + v_B) \times 4$

Solve to get , v_A = 10 m/s and v_B = 30 m/s

Final Answer : (1)

26) **Explaining :** Identify correct options.

Concept :

- Relatiion between x and y is path equation
- $\vec{v} = \frac{d\vec{r}}{dt}$ and $\vec{a} = \frac{d\vec{v}}{dt}$

Solution : x = t and $y = t^2$

Eliminate time t

$\Rightarrow$ $y = x^2$

$\vec{r} = t\hat{i} + t^2\hat{j}$

$\vec{a} = \frac{d^2}{dt^2}(\vec{r}) = 2\hat{j}$

at $t = 0, \vec{r} = \vec{0}$

$t = 1, \vec{r} = 1\hat{i} + 1\hat{j}$

So, displacement = $\sqrt{1^2 + 1^2} = \sqrt{2}$

Final Answer : (4)

27) **Explaining :** Identify which option is correct.

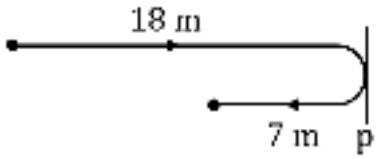
Concept : Distance always increases even if particle reverse the direction.

Solution : This is distance versus time curve and particle is at rest in t = 1 to 2 and t =5 to 6 sec at these moment man may change his direction of motion. But distance will continuous increase so we can't predict that man has changed his direction or not so there are following possibilities.

Case -I: If man is not changing direction.

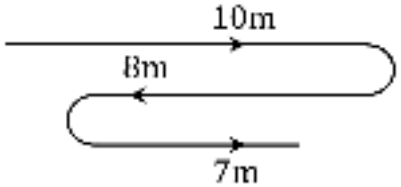
$|\vec{S}| = 25m$

Case-II: If after t = 6 sec he changes his direction but not after t = 2



$|\vec{S}| = 11m$

Case-III: If after t = 2 sec and then t = 6 sec
he changes the direction



$|\vec{S}| = 9m$

Final Answer : (4)

28) **Explaining :** Find maximum x-coordinate.

Concept : $v^2 = u^2 + 2as$

Solution :

$u_x = 3ms^{-1}$, $a_x = -6\ m/s^2$

$v_x^2 = u_x^2 + 2a_x s_x$

$0 = 9 - 2 \times 6 \times s_x$

$s_x = \frac{9}{12} = \frac{3}{4} = 0.75m$

Final Answer : (2)

29) **Explaining :** Find value of v_0 so that v_B will be at angle 30° from verticle.

Concept : Horizontal component of velocity remains constant.

$v^2 = u^2 + 2as$

Solution :

$v_0 \sin 45^\circ = v_B \sin 30^\circ$

$\frac{v_0}{\sqrt{2}} = \frac{v_B}{2} \qquad \therefore v_B = \sqrt{2} v_0$

$v_B^2 \cos^2 30^\circ - v_0^2 \cos^2 45^\circ = 2 \times 9.8 \times 0.1$

$v_0^2 = 1.96$

$v_0 = \sqrt{1.96} = 1.4 m/s$

Final Answer : (2)

30) **Explaining :** Identify the correct option.

Concept : $\vec{A} \cdot \vec{B} = AB \sin\theta$

$\vec{A} \times \vec{B} = AB \sin \hat{n}$

Solution : As per question, $|\vec{P} \cdot \vec{Q}| = \vec{P} \times \vec{Q} = 0$

This is possible if either P = 0 or Q = 0

Magnitude of a null vector is zero.

Final Answer : (1)

31) **Explaining :** Find velocity of wagon.

Concept : Apply COLM

Solution :

$5 \times 10^3 \times 1.2 = (5 \times 10^3 + 10^3)\, v$

$5 \times 10^3 \times 1.2 = 6 \times 10^3\, v$

v = 1 m/s

Final Answer : (3)

32) **Explaining :** Find elongation in spring.

Concept : $F_{net} = ma$

$\left|\vec{F}_{Spring}\right| = kx$

Solution : $\vec{F}_{net} = m\vec{a}$

$mg - kx = ma$

$\Rightarrow 50 - 100x = 5(6)$

$100x = 20$

x = 20 cm

Final Answer : (3)

33) **Explaining :** Find reading of spring balance.

Concept : Spring balance measures the value of tension.

$F_{net} = ma$

Solution : $a = \frac{1 \times 10}{1 + 4} = 2 m/s^2$

$T = 4g \sin 30^\circ - 4a$

$= 4 \times 10 \times \frac{1}{2} - 4 \times 2$

T = 12N

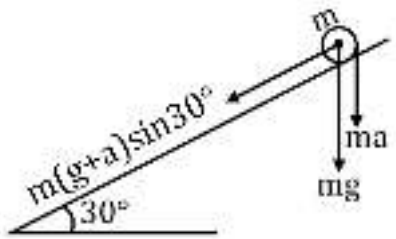
Final Answer : (1)

34) **Explaining :** Find time taken by block to move from top to bottom.

Concept : $g_{eff} = g + a$

$S = ut + \frac{1}{2} at^2$

Solution :



$a' = (g + a) \sin 30^\circ = 6 \text{ m/s}^2$

$s = O + \frac{1}{2} a' t^2$

$t = \sqrt{\frac{2s}{a'}} = \sqrt{\frac{2 \times 12}{(10 + 2) \times \frac{1}{2}}}$

t = 2sec

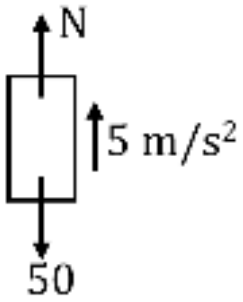
Final Answer : (1)

35) **Explaining :** Find minimum value of F to move the block.

Concept : $f_L = \mu N$

$g_{eff} = g + a$

Solution :



N = m (g + a)

N = 5 (15)

N = 75 N

Now, F = f = μN = 0.2 × 75 = 15 N

Final Answer : (3)

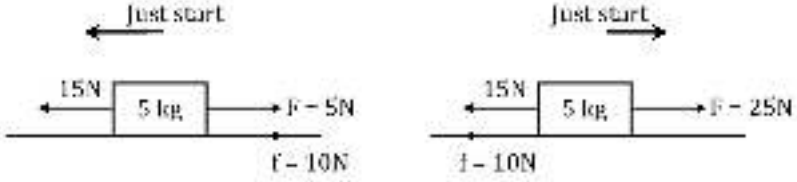
36) **Explaining :** Find maximum and minimum value of force for which block will remain at rest.

Concept :

Solution :

$f_s(max) = 0.2 \times 5 \times 10 = 10N$

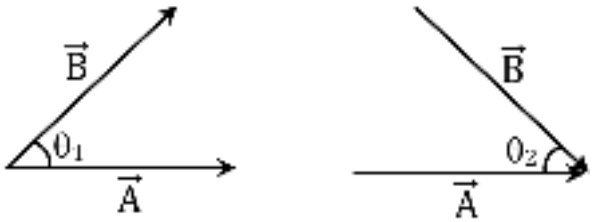
Block will start moving for two values of force as shown



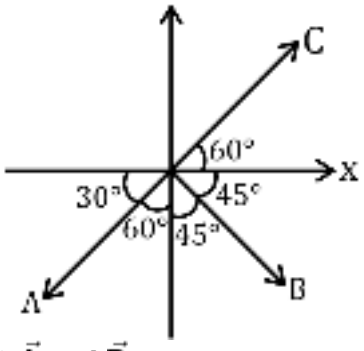
Final Answer : (2)

37) **Explaining :** Find change in momentum.
Concept : $\Delta\vec{p} = \vec{p}_f - \vec{p}_i$
Solution :
$\vec{p}_i = (0.15kg)\left(3\hat{i} + 4\hat{j}\right) ms^{-1}$
$= \left(0.45\hat{i} + 0.6\hat{j}\right) kg\ ms^{-1}$
$\vec{p}_f = (0.15\ kg)\left(-3\hat{i} - 4\hat{j}\right) ms^{-1}$
$= -0.45\hat{i} - 0.6\hat{j}\ kg\ ms^{-1}$
Change in momentum,
$\Delta\vec{p} = \vec{p}_f - \vec{p}_i = \left(-0.45\hat{i} - 0.6\hat{j}\right) - \left(0.45\hat{i} + 0.6\hat{j}\right)$
$= -\left(0.9\hat{i} + 1.2\hat{j}\right)$
Final Answer : (3)

38) **Explaining :** Find angle between given vectors. **Concept :**



Angle = θ_1 Angle = $\vec{\theta}_2$
Solution :



(i) $\vec{A}$ and $\vec{B}$ = 105°
(ii) $\vec{A}$ and $\vec{C}$ = 150°
(iii) $\vec{B}$ and $\vec{C}$ = 105°
Final Answer : (2)

39) **Explaining :** Identity that 4 and 3 are dimensionless or not.
Concept : Principle of homogeniety.
Solution : Dimension of 4t and $3t^2$ will be equal to length L for dimensional correctness.
Hence, $L = 4t = 3t^2$
Dimension of constant 4 and 3 will be $\frac{L}{T}$ and $\frac{L}{T^2}$ respectively.
Final Answer : (3)

40) **Explaining :** Find % error in x.
Concept :

% error in x = $3\left(\frac{4k}{k}\right)+2\left(\frac{\Delta\ell}{\ell}\right)+1\left(\frac{\Delta m}{m}\right)+\frac{1}{2}\left(\frac{\Delta n}{n}\right)$

Solution : A physical quantity $x=\frac{2k^3\ell^2}{m\sqrt{n}}$

Percentage error in

$x=\left(3\times1+2\times2+3\times1+4\times\frac{1}{2}\right)=12\%$

Final Answer : (3)

41) **Explaining :** Find least count of callipers.

Concept : L.C. = 1 MSD – 1 VSD

Solution : 10 MSD = 1 cm

1 MSD = $\frac{1}{10}$ = 0.1 cm

and 10 VSD = 8 MSD

1 VSD = 0.8 MSD

L.C. = 1 MSD – 1 VSD

= 0.2 MSD

= 0.2 × 0.1

= 0.02 cm

Final Answer : (2)

42) **Explaining :** Match the column with column II.

Concept : $y=x\tan\theta\left(1-\frac{x}{R}\right)$

Solution : Compare with standard equation

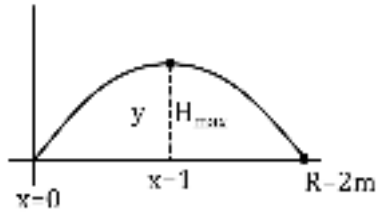
$y=x\tan\theta\left(1-\frac{x}{R}\right)$

to given eq. $y=x\left(1-\frac{x}{2}\right)$

R = 2m and tan θ = 1

θ = 45°

and for H_{max}, put x = 1 (Mid of range) in



$y=x-\frac{x^2}{2}$

$y=\left(1-\frac{1}{2}\right)$

$y=\frac{1}{2}m$

Final Answer : (4)

43) **Explaining :** Find reaction force on hand of gardner.

Concept : $F=v\frac{dm}{dt}$

$\frac{V}{t}$ = Volume flowing per sec

Solution : Rate of flow of water $\frac{V}{t} = \frac{10cm^3}{sec}$

$= 10 \times 10^{-6} m^3/sec$

Density of water $\rho = \frac{10^3 kg}{m^3}$

Cross-sectional area of pipe A = $\pi (0.5 \times 10^{-3})^2$

Force $= m\frac{dv}{dt} = \frac{mv}{t} = \frac{V\rho v}{t} = \frac{\rho V}{t} \times \frac{V}{At} = \left(\frac{V}{t}\right)^2 \frac{\rho}{A} \quad \left(\because v = \frac{V}{At}\right)$

By substituting the value in the above formula we get

F = 0.127 N

Final Answer : (4)

44) **Explaining :** Find ratio of contact force as shown in figures.

Concept : F_{net} = ma

Solution : Acceleration of system in both case is

$a = \frac{F}{3m}$

$\frac{N_1}{N_2} = \frac{\frac{F}{3m} \times 2m}{\frac{F}{3m} \times m} = 2 : 1$

Final Answer : (2)

45) **Explaining :** Find ($\mu_S + \mu_k$).

Concept : $f_L = \mu_S N$

$f_k = \mu kN$

Solution :

$5 = \mu_s g \quad \Rightarrow \mu_s = 0.5$

$4 = \mu_k g \quad \Rightarrow \mu_k = 0.4$

$\Rightarrow \quad \mu_k + \mu_s = 0.9$

Final Answer : (3)

CHEMISTRY

46) **Explaining :** Find atom, molecules, mole

Concept : This question is based on formula based

Solution : Avogadro's law

Final Answer : (3)

47) **Explaining :** Calculate molecules ?

Concept : This question is based on calculate mole and then add molecules

Solution : no. of molecules $= \left(\frac{4.4}{44} \times N_A\right) + \left(\frac{2.24}{22.4} N_A\right)$

Final Answer : (2)

48)

Explaining : Find empirical formula
Concept : This question is based on E.M.P. formula
Solution : -

mole Ratio			
C	54.55/12	4.5	4.5/2.27 ≈ 2
H	9.09/1	9.09	$\frac{9.09}{2.27} \approx 4$
O	36.36/16	2.27	$\frac{2.27}{2.27} = 1$

C_2H_4O = Empirical formula.
Final Answer : (4)

49) **Explaining :** Find no. of atoms
Concept : This question is based on atom = mole × N_A × atomicity
Solution :
(1) $\frac{28}{28} \times 2$ = 2 mole atoms
(2) $\frac{44.8}{22.4} \times 2$ = 4 mole atoms
(3) $\frac{18}{18} \times 3$ = 3 mole atoms
(4) 1.5 mole atoms
Final Answer : (2)

50) **Explaining :** minimum molecular mass
Concept : This question is based on percentage composition
Solution :
% of O = $\frac{1 \times \text{At.wt of 'O'}}{M_w} \times 100$
$3.2 = \frac{1 \times 16}{M_w} \times 100 \Rightarrow M_w = 500$
Final Answer : (1)

51) **Explaining :** Find L.R.
Concept : This question is based on L.R.
Solution : $A_2 + 2B \rightarrow 2AB$

$x = \frac{4}{1} \quad \frac{6}{2} = 3$

(L.R.)

Final Answer : (2)

52)

Explaining : Find no. of lines in UV region.
Concept : This question is based on H spectrum.
Solution :
$5 \rightarrow 1$
$4 \rightarrow 1$
$3 \rightarrow 1$
$2 \rightarrow 1$
Final Answer : (2)

53) **Explaining :** Find correct.
Concept : This question is based on H spectrum.
Solution :
$E_2 - E_1 = 10.2$ eV
$E_3 - E_2 = 1.89$ eV
$E_4 - E_3 = 0.66$ eV
Final Answer : (1)

54) **Explaining :** Find shortest wavelength.
Concept : This question is based on H spectrum.
Solution :
Shortest wavelength $\infty \rightarrow 3$

$$\frac{1}{\lambda} = R(3)^2 \left(\frac{1}{(3)^2} - \frac{1}{(\infty)^2} \right)$$

$$\frac{1}{\lambda} = R(9) \times \frac{1}{9} = R$$

$$\lambda = \frac{1}{R}$$

Final Answer : (3)

55) **Explaining :** Find excited state.
Concept : This question is based on Energy Bohr question.
Solution :
P.E. = 2 × T.E.

$$\text{T.E.} = \frac{\text{P.E.}}{2} = \frac{-6.8}{2} = -3.4\,\text{eV}$$

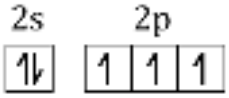
Final Answer : (1)

56)

Explaining : The orbital diagram in which the Aufbau principle is violated is :
Concept : This question is based on, firstly lower energy level filled, then electrons move to higher energy level (Aufbau principle)

In option - (1) (Hund's rule violation)
(2) Aufbau principle (2s filled before 2p)

Solution :



Final Answer : (2)

57)

Explaining : Definition
Concept : This question is based on-
Pauli's Rule – No two electrons in an atom can have the same set of four quantum numbers
Aufbau Rule – In the ground state of atoms, orbitals are filled in the order of their increasing energies.
Hund's Rule – Pairing of electrons in the Principle orbitals belonging to the same subshell does not take place until each orbital is singly occupied.
Heisenberg's principle – It is impossible to determine the exact position and exact momentum of a subatomic particle simultaneously.
Solution : A – (r), B – (t), C – (p), D – (s)
Final Answer : (3)

58)

Explaining : The equilibrium constant (K) of the reaction
Concept : This question is based on multiple reaction
Solution :

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ $\left(\frac{1}{K_1}\right)$

$3H_2(g) + \frac{3}{2} O_2(g) \rightleftharpoons 3H_2O(g)$ $\left(K_3^3\right)$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (K_2)

$2NH_3(g) + \frac{5}{2} O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$ $K = \frac{K_2K_3^3}{K_1}$

Final Answer : (1)

59)

Explaining : For the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, the value of K_c at 800° C is 0.1. When the equilibrium concentrations of both the reactants is 0.5 mol, what is the value of K_p at the same temperature
Concept : This question is based on $K_p = K_c(RT)^{\Delta n_g}$
Solution :
$K_p = K_c(RT)^{\Delta n_g}$
Since, $\triangle n_g = 0$, hence, $K_p = K_c$, = 0.1]

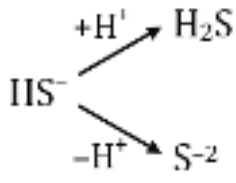
Final Answer : (2)

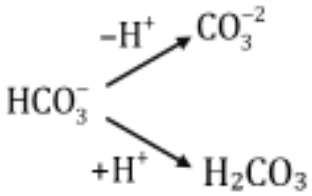
60)

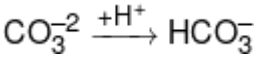
Explaining : Calculate K_p.
Concept : This question is based on K_p & K_c.
Solution :
$\Delta n_g = 1 - 2 = -1$
$\therefore K_p = K_c(RT)^{\Delta n_g}$
$K_p = K_c(RT)^{-1}$
Final Answer : (4)

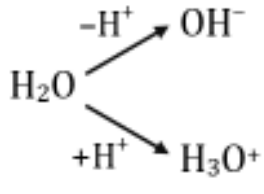
61)

Explaining : Which of the following cannot act as both Bronsted acid and base.
Concept : This question is based on Bronsted acid base concept.
Solution :

(1) 

(2) 

(3) 

(4) 

Final Answer : (2)

62)

Explaining : Solubility product constants (K_{sp}) of salts of types MX, MX_2 at M_3X at temperature T are 4.0×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15} respectively. Solubility (mol dm^{-3}) of the salt at temperature T are in the order
Concept : This question is based on Calculation of solubility.
Solution :

$$(MX) : S = \sqrt{K_{sp}} = \sqrt{4 \times 10^{-8}} = 2 \times 10^{-4}$$

$$(MX_2) : S = \left(\frac{1}{4} \times 3.2 \times 10^{-14}\right)^{1/3} = 2 \times 10^{-5}$$

$$(M_3X) : S = \left(\frac{1}{27} \times 2.7 \times 10^{-15}\right)^{1/4} = 10^{-4}$$

$MX_2 < M_3X < MX$

Final Answer : (4)

63)

Explaining : What is the minimum volume of water required to dissolve 1 g of calcium sulphate at 298 K. K_{sp} for $CaSO_4$ is 9.0×10^{-6}.

Concept : This question is based on Solubility product.

Solution :

$$CaSO_4 \rightleftharpoons Ca^{+2} + SO_4^{-2}$$

$K_{sp} = S^2 \Rightarrow S = (9 \times 10^{-6})^{1/2}$

$S = 3 \times 10^{-3}$

$$3 \times 10^{-3} = \frac{w/M_w}{V}$$

$$V = \frac{1/136}{3 \times 10^{-3}} = 2.45 \text{ L}$$

Final Answer : (1)

64)

Explaining : The ratio of volumes of CH_3COOH, 0.1 N to CH_3COONa, 0.1 N required to prepare a buffer solution of pH 5.74 is (given : pK_a of CH_3COOH = 4.74)

Concept : This question is based on pH of acidic Buffer.

Solution :

$$pH = pK_a + \log \frac{(N \times V)_{salt}}{(N \times V)_{acid}}$$

$$5.74 = 4.74 + \log \frac{(0.1 \times V)_{salt}}{(0.1 \times V)_{acid}}$$

$$\frac{V_{Salt}}{V_{acid}} = \frac{10}{1}$$

$$\frac{V_{CH_3COOH}}{V_{CH_3COONa}} = \frac{1}{10}$$

Final Answer : (4)

65)

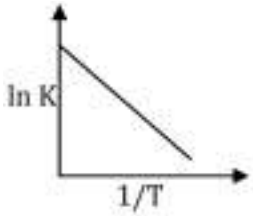
Explaining : The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of first order reaction is -5×10^3 K. The value of E_a of the reaction is.

Concept : This question is based on Arrhenius equation

Solution :

$$\ln K = \ln A - \frac{E_a}{R} \times \frac{1}{T} \qquad \text{slope}(m) = \frac{-E_a}{R}$$

$y = C + mx$ Intercept (C) = ln A



Given : slope = $\frac{-E_a}{R} = -5 \times 10^3$

$E_a = 5 \times 10^3 R = 5 \times 10^3 \times 8.314$

$E_a = 41.5 \times 10^3$ J mol^{-1}

$E_a = 41.5$ kJ mol^{-1}

Final Answer : (1)

66) **Explaining :** For the reaction : $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ rate of appearance of NH_3 is 6.8×10^{-3} g/min, then rate of disappearance of $H_2(g)$ at the same condition will be :

Concept : This question is based on $ROR = \frac{(ROD)H_2}{3} = \frac{(ROA)NH_3}{2}$

Solution : $\frac{1}{2} \times \frac{6.8 \times 10^{-3}}{17} \times 2 \times 3$

Final Answer : (4)

67) **Explaining :** Which of the following statements regarding rate constant is correct?

Concept : This question is based on rate constant.

Solution : Rate constant is temperature dependent.

Final Answer : (2)

68) **Explaining :** Rate of reaction.

Concept : This question is based on Rate.

Solution :

$[0.10]^a [0.10]^b = 0.030$ ...(1)

$[0.20]^a [0.10]^b = 0.060$ ...(2)

$[0.10]^a [0.050]^b = 0.015$ ...(3)

(2) / (1) a = 1

(1) / (2) b = 1

Rate = k[IC□] $[H_2]$

Final Answer : (1)

69) **Explaining :** In a zero order reaction half life is 100 sec. After how much time 7/8 fraction of reactant will be reacted?

Concept : This question is based on zero order reaction.

Solution : $x = Kt \quad \Rightarrow \quad \frac{7a}{8} = \frac{a}{200} \times t \qquad \Rightarrow t = 175\,\text{sec}$

Final Answer : (3)

70) **Explaining :** Consider the Arrhenius equation given below and mark the **CORRECT** option, $k = Ae^{-E_a/RT}$

Concept : This question is based on Arrhenius equation.

Solution : Rate constant increases exponentially with decreasing activation energy and increasing temperature.
Final Answer : (4)

71) **Explaining :** Which of the following is correctly matched?
Concept : This question is based on nature of oxides
Solution :
SO_3 - acidic
CO_2 - acidic
NO_2 - acidic
ZnO - amphoteric
Final Answer : (4)

72) **Explaining :** Among the following species, select the diamagnetic molecule:
Concept : This question is based on application of MOT
Solution :
B_2, n = 2 CO, n = 0
NO, n = 1 O_2, n = 2
where n is no. of unpaired electrons.
Final Answer : (2)

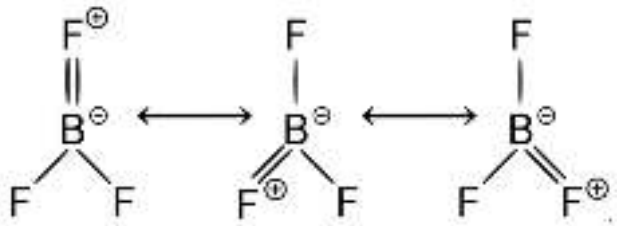
73) **Explaining :** For SO_2, NO_2^- and N_3^-, identify the species in which the central atom is sp-hybridized:
Concept : This question is based on hybridization
Solution :
$SO_2 \rightarrow sp^2$
$NO_2^- \rightarrow sp^2$
$N_3^- \rightarrow sp$
Final Answer : (3)

74) **Explaining :** Analyse both the statements
Concept : This question is based on reactivity of elements and nature of oxides.
Solution :
In a period , reactivity of metals decrease moving left to right while, reactivity of non-metals increases on moving left to right.
Metallic oxides – Basic in nature
Non-metallic oxides – Acidic in nature.
Final Answer : (3)

75) **Explaining :** Correct statement will be :
Concept : This question is based on multiple concept
Solution :

$$\text{Bond order} = \frac{\text{Number of bonds present between atoms}}{\text{Total number of resonance structures}}$$

$\because$ has three resonance structures.



So, bond order $= \frac{4}{3} = 1\frac{1}{3}$

Electron gain enthalpy of oxygen will be least negative in their group.

N_2O will be example of sub-oxide

In the see-saw geometry of SF_4, lone pair occupies equatorial position

Final Answer : (2)

76) **Explaining :** The maximum ionization energy would be associated with the electronic configuration :

Concept : This question is based on ionization energy

Solution :

Ionization energy increases along the period and decreases down the group.

Ionization energy of noble gases is maximum.

Final Answer : (3)

77) **Explaining :** The correct order of increasing metallic character among Si, Be, Mg, Na, P is :

Concept : This question is based on metallic character

Solution :

P < Si < Be < Mg < Na

Metallic character $\propto \frac{1}{\text{I.E.}}$

Final Answer : (3)

78) **Explaining :** In which of the following arrangements the order is not according to the property indicated against it?

Concept : This question is based on multiple concept

Solution : B < C < O < N – Increasing first ionization enthalpy

Final Answer : (2)

79) **Explaining :** Arrange the following in increasing order of covalent character - NaCl, $MgCl_2$, $AlCl_3$

Concept : This question is based on polarization

Solution :

Cation size is decreasing in the order: $Na^+ > Mg^{2+} > Al^{3+}$

The increasing order of covalent character is $NaCl < MgCl_2 < AlCl_3$

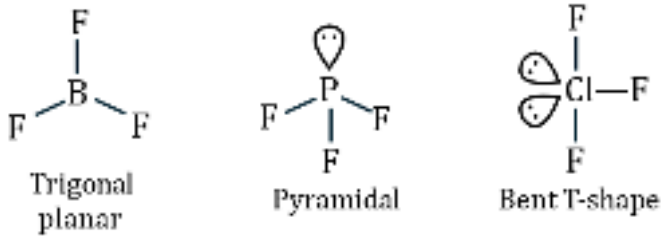
Final Answer : (1)

80) **Explaining :** The ionic solids JX, LY and MZ are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first ?

Concept : This question is based on lattice energy

Solution :
The highest lattice energy is given by one with the highest ionic charged and smallest ionic sizes.
Final Answer : (4)

81) **Explaining :** The corrects increasing order for bond angles among BF_3, PF_3 and ClF_3 is :
Concept : This question is based on bond angle
Solution :
$ClF_3 < PF_3 < BF_3$



Final Answer : (3)

82) **Explaining :** Arrange the noble gases in increasing polarizability : He, Ne, Ar, Xe
Concept : This question is based on polarizability
Solution :
He < Ne < Ar < Xe
Polarizability increase as atomic radii increase and more diffuse electron cloud follow for greater distortion.
Final Answer : (1)

83) **Explaining :** Which molecule is non-polar?
Concept : This question is based on polarity of moelcules.
Solution : CO_2 is linear and symmetrical.
Final Answer : (4)

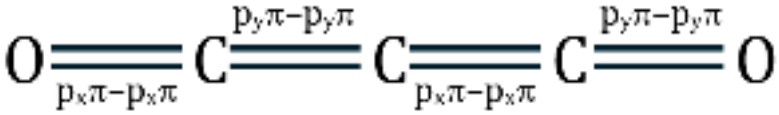
84) **Explaining :** Which ion is paramagnetic?
Concept : This question is based on magnetic behaviour of ions.
Solution : Fe^{3+} contains unpaired electrons.
Final Answer : (4)

85) **Explaining :** Which element has smallest atomic radius
Concept : This question is based on atomic radius
Solution : Atomic radius decreases across period.
Final Answer : (4)

86) **Explaining :** Where a, b, c, d are elements of group-17 if difference between atomic number of elements d and b is x and difference between atomic number of elements c and a is y. What is the value of x-y.
Concept : This question is based on electron affinity
Solution :
Electron affinity order : Cl > F > Br > I

a-F, b-Cl, c-Br, d-I
x = 53 – 17 = 36
y = 35 – 9 = 26
x – y = 10
Final Answer : (1)

87) **Explaining :** In C_3O_2 molecule z-axis is assumed as molecular axis. The maximum no. π-bonds present in x-z plane.
Concept : This question is based on overlapping of orbitals and type of bonds.
Solution :



Final Answer : (2)

88) **Explaining :** PCl_5 is well known, but NCl_5 is not.
Concept : This question is based on existence and non-existence of molecules.
Solution : Nitrogen doesn't have d-orbitals in it valence shell.
Final Answer : (2)

89) **Explaining :** Which compound shows hydrogen bonding?
Concept : This question is based on hydrogen bonding.
Solution : HF forms strong hydrogen bonds.
Final Answer : (2)

90) **Explaining :** Identify the incorrect match:
Concept : This question is based on IUPAC naming of elements with atomic number > 100
Solution :
103 – Lawrencium
112 – Copernicium
Final Answer : (4)

BIOLOGY

91) **Solution :** Ashwagandha and Belladonna are medicinal plants of the family Solanaceae.
Final Answer : (3)

92) **Solution :** This formula indicates papilionaceous petals and diadelphous stamens, classic for Fabaceae.
Final Answer : (1)

93) **Solution :** Stamens united into two bundles (diadelphous) is characteristic of pea flowers (Fabaceae).
Final Answer : (3)

94) **Solution :** The cells in the region of elongation undergo rapid elongation and enlargement, being responsible for length growth.
Final Answer : (3)

95) **Solution :** Stamens attached to the perianth (lily) are epiphyllous.
Final Answer : (2)

96) **Solution :** It is cymose type of inflorescence in which flowers are borne in a basipetal order
Final Answer : (2)

97) **Solution :** Stem develops from plumule of embryo.
Final Answer : (3)

98) **Solution :** All the given examples are underground stem modification.
Final Answer : (4)

99) **Solution : Problem Statement:** The question asks to identify the type of phyllotaxy (the arrangement of leaves on a stem) present in three different plants: Mustard (1 leaf arises from each node), Guava (2 leaf arises from each node), and *Alstonia* (More then 2 leaf arises from each node), respectively.
Final Answer : (2)

100) **Solution :** Axillary bud is derived from shoot apical meristem.
Final Answer : (4)

101) **Explain Question:**
The question asks for the tissue system category to which leaf mesophyll cells belong.
Concept :
This question is based on Ground tissue system.
Explanation:
The mesophyll in the leaf is made of parenchyma cells which belong to the ground tissue system.
Final answer : 2

102) **Explain Question :** The question is asking to identify a mature, differentiated plant cell type that contains cytoplasm but lacks a nucleus.
Concept : This question is based on Complex Permanent Tissue.
Explanation: 'The sieve tube elements possess cytoplasm but lack a nucleus.'
Final answer : (4)

103) **Question Explanation:** Identify the arrangement of primary xylem.
Concept: Exarch xylem.
Solution: The figure shows the vascular tissue in a root. The smaller, younger protoxylem elements are located towards the periphery, while the larger, mature metaxylem elements are

towards the center. This specific arrangement of primary xylem is called exarch, which is characteristic of roots.
Final Answer : (1)

104) **Solution :** In monocotyledon plant vascular bundles is closed type because vascular bundles have no cambium and they do not form secondary xylem and phloem.
Final Answer : (2)

105) **Solution : A -** Trichomes; **B -** Hypodermis; **C -** Vascular Bundles.
Final Answer : (3)

106) **Solution :** This floral formula belongs to family solanaceae and tobacco is a member of solanaceae.
Final Answer : (1)

107) **Solution :** Pulses yielding main family of plants is papilionaceae (Fabaceae).
Final Answer : (4)

108) **Solution :** P→c, Q→a, R→b, S→d

Column-I		**Column-II**	
(P)	Tetradynamous	(c)	Cruciferae
(Q)	Diadelphous	(a)	Leguminosae
(R)	Monoadelphous	(b)	Malvaceae
(S)	Syngenesious	(d)	Asteraceae

Final Answer : (1)

109) **Solution :** Only E statement is correct about sclerenchyma. It provides mechanical support to the growing aerial parts of the monocot plant.
Final Answer : (1)

110) **Solution :** i, iii, iv
(i) Growth in plants is largely restricted to specialized regions.
(iii) Ground tissue of leaves is known as mesophyll.
(iv) Dicot roots show secondary growth.
Final Answer : (3)

111) **Solution :** *Triticum* and *Mangifera* belong to same division (Angiospermae) and kingdom (Plantae).
Final Answer : (1)

112) **Solution : A -** *Datura*; **B -** *Felis*; **C -** Felidae.

Final Answer : (2)

113) **Solution :** Consciousness is considered a defining property of living organisms because all organisms are aware of their surroundings and respond to external environmental stimuli.
Final Answer : (3)

114) **Solution :** Lichens are the best indicators of air pollution.
Final Answer : (3)

115) **Solution :** Biological concept of species is neither followed by *Acetobacter* and *Mycoplasma* because they are prokaryotes and nor by mule because it is a sterile animal.
Final Answer : (1)

116) **Solution :** Diatoms floats passively in water while dinoflagellates have 2 flagella for floating.
Final Answer : (3)

117) **Explain Question :** Analyze the table and figure out the missing values A and B based on Whittaker's Five Kingdom Classification:
Concept : This question is based on Five Kingdom Classification.
Solution : The correct answer is Option (2): (A) = Multicellular / loose tissue, (B) = Present is some members
Final Answer : (2)

118) **Explain Question:** Question asking which character not used by R.H. Whittaker in his five kingdom classification. **Concept:** Five Kingdom Classification-R.H. Whittaker
Solution: In Five Kingdom Classification proposed by R.H. Whittaker (1969).The main criteria for classification used by him include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.
-Type of spore formation not used in Five Kingdom Classification proposed by R.H. Whittaker
Final answer : (3)

119) **Solution :** Diatoms floats passively in water and they don't have flagella.
Final Answer : (3)

120) **Question Explanation :** True regarding thermoacidophiles
Concept : Archaebacteria
Solution : b. They are members of archaea - correct.
c. They are found in certain hydrothermal springs - correct.
e. Their enzyme activate at acidic pH - correct.
Final Answer : (2)

121) **Solution :** All the given statements are correct for *Mycoplasma*.
Final Answer : (4)

122) **Solution :** In 5 kingdom classification, Blue green algae (Cyanobacteria) are placed in Monera.
Final Answer : (3)

123) **Solution :** *Puccinia*, *Ustilago* and *Agaricus* belong to basidiomycetes and *Aspergillus* belongs to Ascomycetes.
Final Answer : (3)

124) **Solution :** Image given in 1 option depicts slime mould.
Final Answer : (1)

125) **Solution :** In homosporous pteridophytes, gametophytic generation is called prothallus, which is small but multicellular and mostly photosynthetic.
Final Answer : (3)

126) **Solution : X –** Mid rib, **Y –** Daughter colony, **Z -** Air bladder
Final Answer : (2)

127) **Solution :** Protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore. **Final Answer : (3)**

128) **Solution :** Double fertilization and presence of companion cells are exclusive features of angiosperms.
Final Answer : (3)

129) **Solution :** In the life cycle of mosses, the gametophyte has two stages (A and B). These stages can be called **A -** Protonema; **B -** Leafy stage.
Final Answer : (1)

130)

Explain Question : The question asks to identify three specific structures, marked A, B, and C, on a provided diagram of a bryophytic plant i.e. *Marchantia*.
Concept : This question is based on Liverworts
Explanation : A-Archegoniophore, B-Gemma cup, and C-Unicellular rhizoids.
Final answer : (4)

131) **Explanation :** The correct developmental sequence of spermatogenesis is Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa.
Final Answer : (1)

132) **Explanation :** Menstruation does NOT occur after fertilization.
Final Answer : (2)

133) **Explanation:** Both statements correct; reason explains assertion.
Final Answer : (1)

134) **Explanation :** Prostate does not produce ovum.
Final Answer : (2)

135) **Explanation:** Uterus opens into vagina through narrow cervix.
Final Answer : (3)

136) **Explanation:** Both statement A and statement B are correct.
Final Answer : (2)

137) **Explanation:** A-1, B-2, C-3

A. 1st month	1. Heart formed
B. 2nd month	2. Limbs develop
C. 3rd month	3. Major organs formed

Final Answer : (4)

138) **Explanation:** Oxytocin induces uterine contraction.
Final Answer : (3)

139) **Explanation:** Prolactin produces milk; oxytocin releases it.
Final Answer : (1)

140) **Explanation:** Head has DNA, acrosome has enzymes, tail helps in movement.
Final Answer : (3)

141) **Explanation:** Reproductive health focuses on the healthy functioning of the reproductive system and the physical, mental and social well being related to reproduction.
Final Answer : (1)

142) **Explanation:** Coitus interruptus is not fully reliable.
Final Answer : (2)

143) **Explanation:** All barrier methods prevent sperm entry.
Final Answer : (4)

144) **Explanation:** Vasectomy = male sterilization, Tubectomy = female sterilization.
Final Answer : (1)

145) **Explanation:** MTP has risks if not done properly.

Final Answer : (2)

146) **Explanation:** STIs are preventable.
Final Answer : (3)

147) **Explanation:** IVF occurs in lab conditions.
Final Answer : (3)

148) **Explanation:** IUD can be copper releasing or hormonal.
Final Answer : (1)

149) **Explanation:** They prevent ovulation.
Final Answer : (1)

150) **Explanation:** Condoms are effective preventive method.
Final Answer : (1)

151) **Explanation:** A-1, B-2, C-3, D-4

A. Porifera	1. Pores all over body
B. Cnidaria	2. Radial symmetry
C. Arthropoda	3. Jointed appendages
D. Echinodermata	4. Water vascular system

Final Answer : (3)

152) **Explanation:** A-1, B-2, C-3, D-4

A. Pisces	1. Gills
B. Amphibia	2. Dual habitat
C. Reptilia	3. Dry scaly skin
D. Aves	4. Feathers

Final Answer : (1)

153) **Explanation :** A-1, B-2, C-3

Column I	Column II
A. Asymmetry	1. Sponges
B. Radial symmetry	2. Jellyfish
C. Bilateral symmetry	3. Humans

Final Answer : (3)

154) **Explanation:** Segmentation is not present in all non-chordates. Some non-chordates have closed circulatory system (Annelida).
Final Answer : (1)

155) **Explanation:** Dorsal nerve cord is present in chordates.
Final Answer : (1)

156) **Explanation:** Ostia define pore-bearing nature.
Final Answer : (1)

157) **Explanation:** Ostia define pore-bearing nature.
Final Answer : (1)

158) **Explanation:** Arthropoda includes insects, crabs etc.
Final Answer : (3)

159) **Explanation:** Whale and bat are mammals.
Final Answer : (2)

160) **Explanation:** Adults show radial symmetry.
Final Answer : (3)

161) **Explanation :** A-1, B-2, C-3, D-4

A. Skin	1. Respiration
B. Liver	2. Bile secretion
C. Heart	3. Circulation
D. Kidney	4. Excretion

Final Answer : (4)

162) **Explanation:** Trachea for respiration, Malpighian tubules for excretion, hemolymph for circulation.
Final Answer : (2)

163) **Explanation:** Frog = amphibian; cockroach = arthropod.
Final Answer : (1)

164) **Explanation:** Frog also respires through skin.
Final Answer : (3)

165) **Explanation:** Cockroach uses trachea, not gills.

Final Answer : (2)

166) **Explanation:** Adult frog is ureotelic, not ammonotelic.
Final Answer : (3)

167) **Explanation:** Frog is amphibian.
Final Answer : (1)

168) **Explanation:** Tracheal system present.
Final Answer : (2)

169) **Explanation:** Two atria + one ventricle.
Final Answer : (2)

170) **Explanation:** Open circulation means no closed vessels.
Final Answer : (1)

171) **Solution :** Double fertilization is unique to angiosperms in which one male gamete fuses with egg and the other with secondary nucleus.
Final Answer : (1)

172) **Solution :** Synergids help in pollen tube entry into embryo sac with the help of filiform apparatus.
Final Answer : (1)

173) **Solution :** The correct statements are:
a. Tapetum provides nourishment to developing pollen grains → **Correct**
b. Endothecium helps in dehiscence of anther → **Correct**
c. Middle layers persist in mature anther → **Incorrect (they usually degenerate early)**
d. Microspore tetrads are formed after meiosis → **Correct**
Therefore, the correct option is: **(1) a, b and d only**
Final Answer : (1)

174) **Solution :** Check each statement:
a. Egg apparatus consists of one egg cell and two synergids → **Correct**
b. Antipodals are present near micropyle **→ Incorrect (they are present at the chalazal end)**
c. Central cell contains two polar nuclei → **Correct**
d. Embryo sac is generally 8-nucleate and 7-celled → **Correct**
Therefore, the incorrect statement is **b only**
Final Answer : (1)

175) **Solution :**
a. Xenogamy brings genetically different pollen grains → **Correct**

b. Geitonogamy is genetically similar to autogamy → **Correct**
c. Autogamy requires external pollinating agents → **Incorrect**
(autogamy occurs within the same flower and generally does not require external agents)
d. Xenogamy increases variation → **Correct**
Hence, the correct statements are a, b and d.
Final Answer : (1)

176) **Solution :**
a. Apomixis produces seeds without fertilization → **Correct**
b. Polyembryony means many embryos in one seed → **Correct**
c. Adventive embryony develops from nucellus → **Correct**
d. Hybrid varieties cannot retain traits generation after generation if they produce apomictic seeds → **Incorrect**
In fact, apomictic seeds help hybrids retain parental traits generation after generation.
Therefore, correct statements are 1, 2 and 3 only.
Final Answer : (1)

177) **Solution :**
a. Persistent nucellus in seed is called perisperm → **Correct**
b. Coconut water is free nuclear endosperm → **Correct**
c. Albuminous seeds lack endosperm → **Incorrect** (albuminous seeds retain endosperm)
d. Aleurone layer is protein rich → **Correct**
Therefore, the correct option is: 1. a, b and d only
Final Answer : (1)

178) **Solution :** A - i, B - ii, C - iii, D - iv

A. Micropyle	i. Pollen tube's entry
B. Chalaza	ii. Basal part of ovule
C. Funicle	iii. Attachment stalk
D. Hilum	iv. Scar on seed

Final Answer : (3)

179) **Solution :** A - ii, B - i, C - iii, D - iv

A. *Hydrilla*	ii. Water pollination
B. *Amorphophallus*	i. Insect pollination
C. Coleoptile	iii. Protects plumule
D. Coleorhiza	iv. Protects radicle

Final Answer : (4)

180) **Solution :**
Cleistogamy is found in plants like *Viola*, *Oxalis* and *Commelina*.
Sunhemp does not show cleistogamy
Final Answer : (3)